LAB MID-TERM PERPARTION

EMBEDDED SYSTEMS

Experiment # 01

Introduction to Edsim51 simulator with demo examples

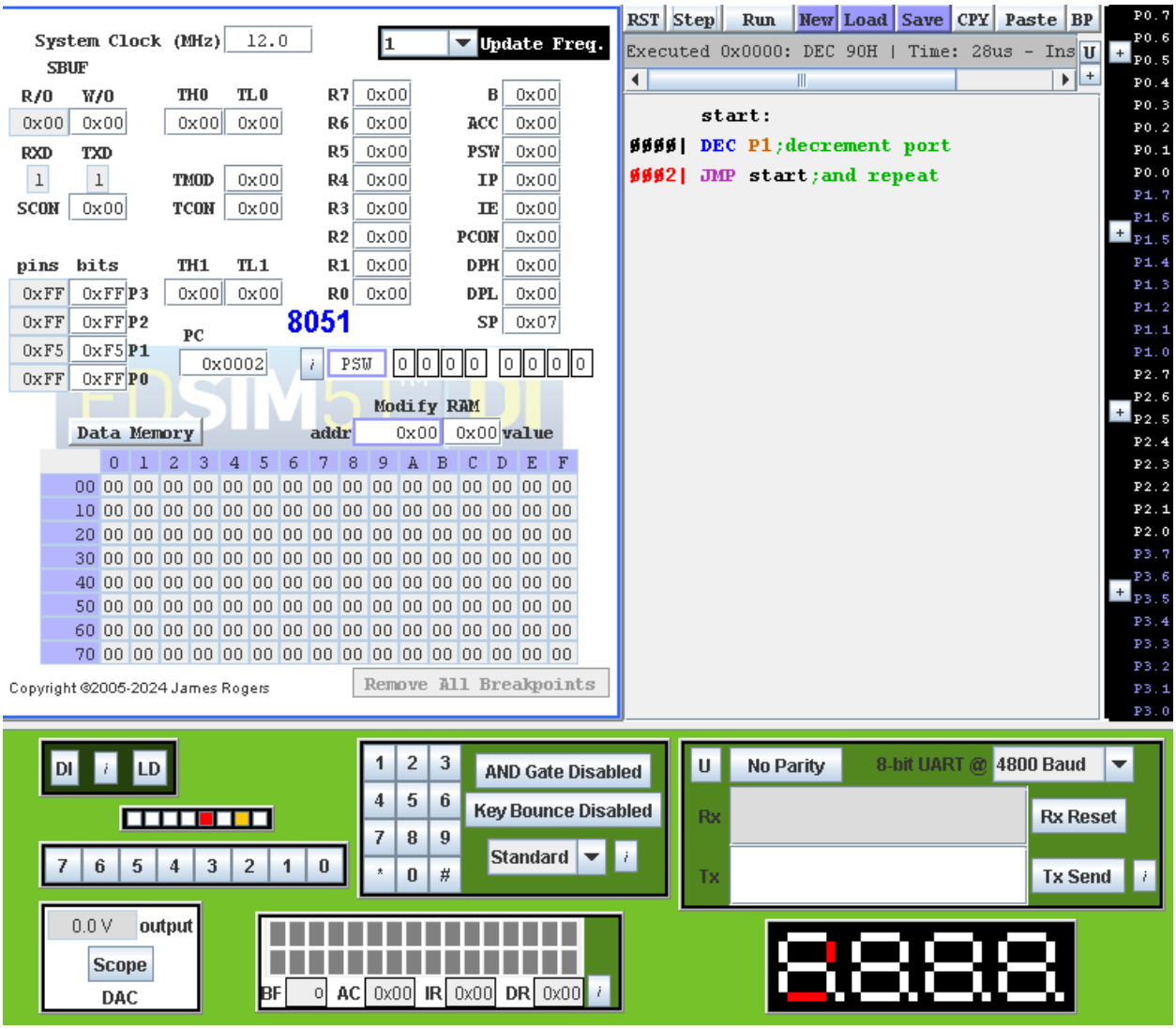
TASK #1:

This program displays the binary pattern from 0 to 255 (and back to 0) on the LEDs interfaced with port 1. A 1 in the pattern is represented by the LED on, while a 0 in the pattern is represented by the LED off. However, logic 0 on a port 1 pin turns on the LED, therefore it is necessary to write the inverse of the pattern to the LEDs. The easiest way to do this is to send the data FFH to 0 (and back to FFH) to the LEDs Since port 1 is initially at FFH all we need to do is continuously decrement port 1.

CODE:

```
start:

DEC P1;decrement port

JMP start;and repeat
```

OUTPUT:



TASK #2:

This program very simply echoes the switches on P2 to the LEDs on P1. When a switch is closed a logic 0 appears on that P2 pin, which is then copied to that P1 bit which turns on that LED. Therefore, a closed switch is seen as a lit LED and vice versa.

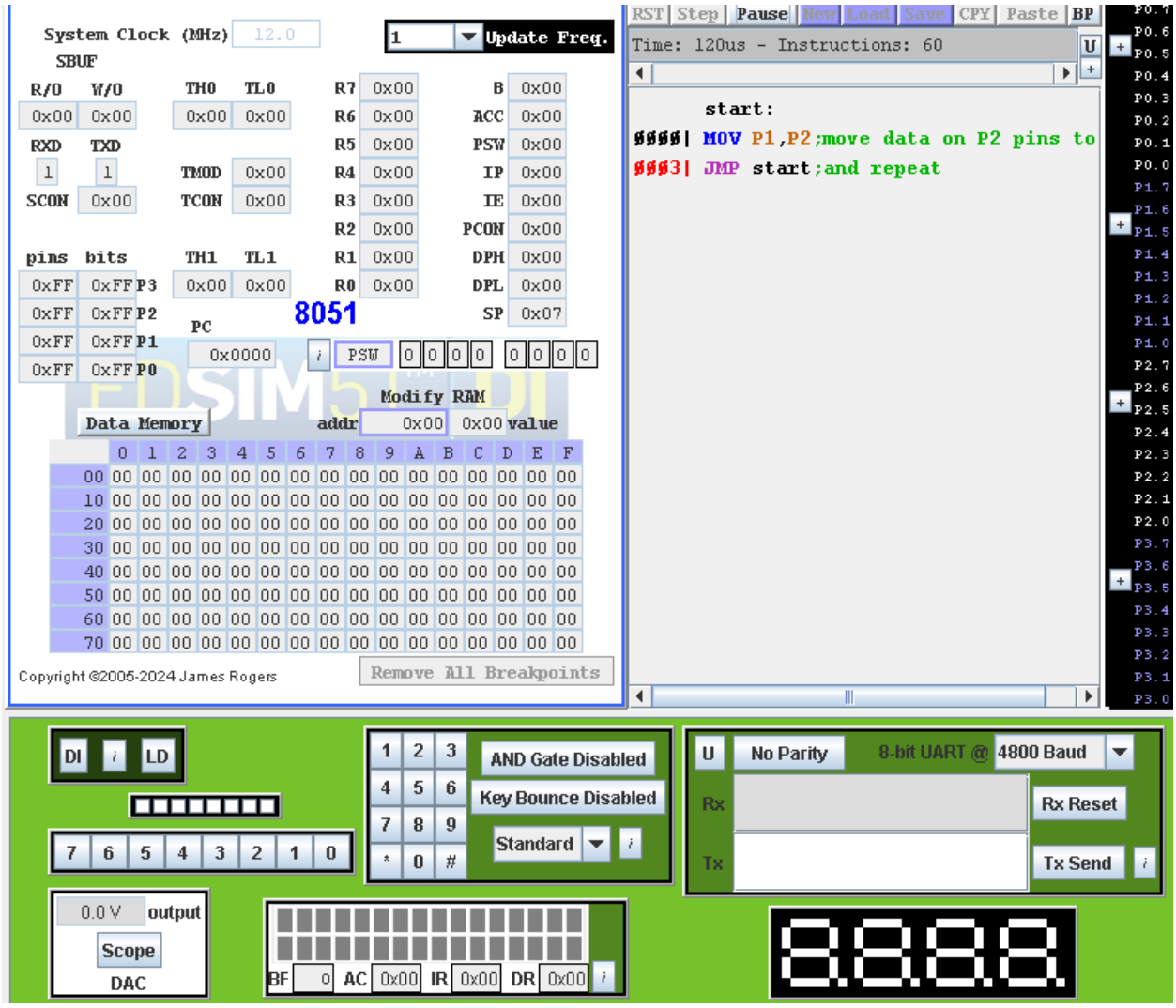
CODE:

start:

MOV P1,P2;move data on P2 pins to P1

JMP start;and repeat

OUTPUT:



TASK #3:

Interface an Analog-to-Digital Converter (ADC) with the 8051 to read an analog signal and display its digital equivalent.

CODE:

ORG 0000H

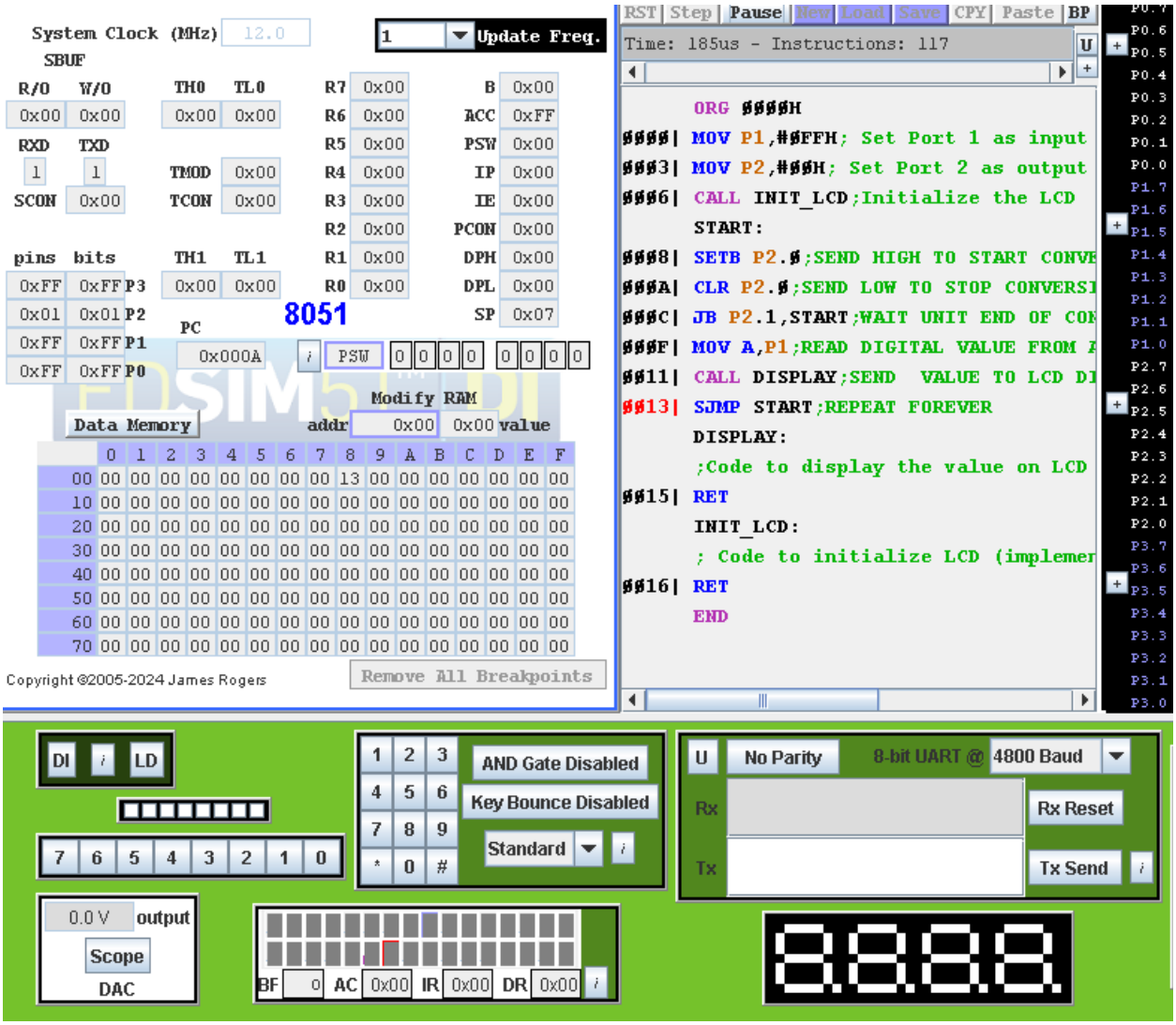
MOV P1,#0FFH; Set Port 1 as input port (for ADC data)

MOV P2,#00H; Set Port 2 as output port (for control signals)

MOV DPTR,#LCD_CMD; Initialize LCD Command Pointer

```
CALL INIT_LCD;Initialize the LCD
START:
SETB P2.0;SEND HIGH TO START CONVERSION(SC PIN)
CLR P2.0;SEND LOW TO STOP CONVERSION
JB P2.1,START;WAIT UNIT END OF CONVERSION(EOC PIN)
MOV A,P1;READ DIGITAL VALUE FROM ADC(8-BIT)
CALL DISPLAY;SEND  VALUE TO LCD DISPLAY
SJMP START;REPEAT FOREVER
DISPLAY:
;Code to display the value on LCD (implement your own LCD routine)
RET
INIT_LCD:
; Code to initialize LCD (implement your own LCD routine)
RET
END
```

OUTPUT:



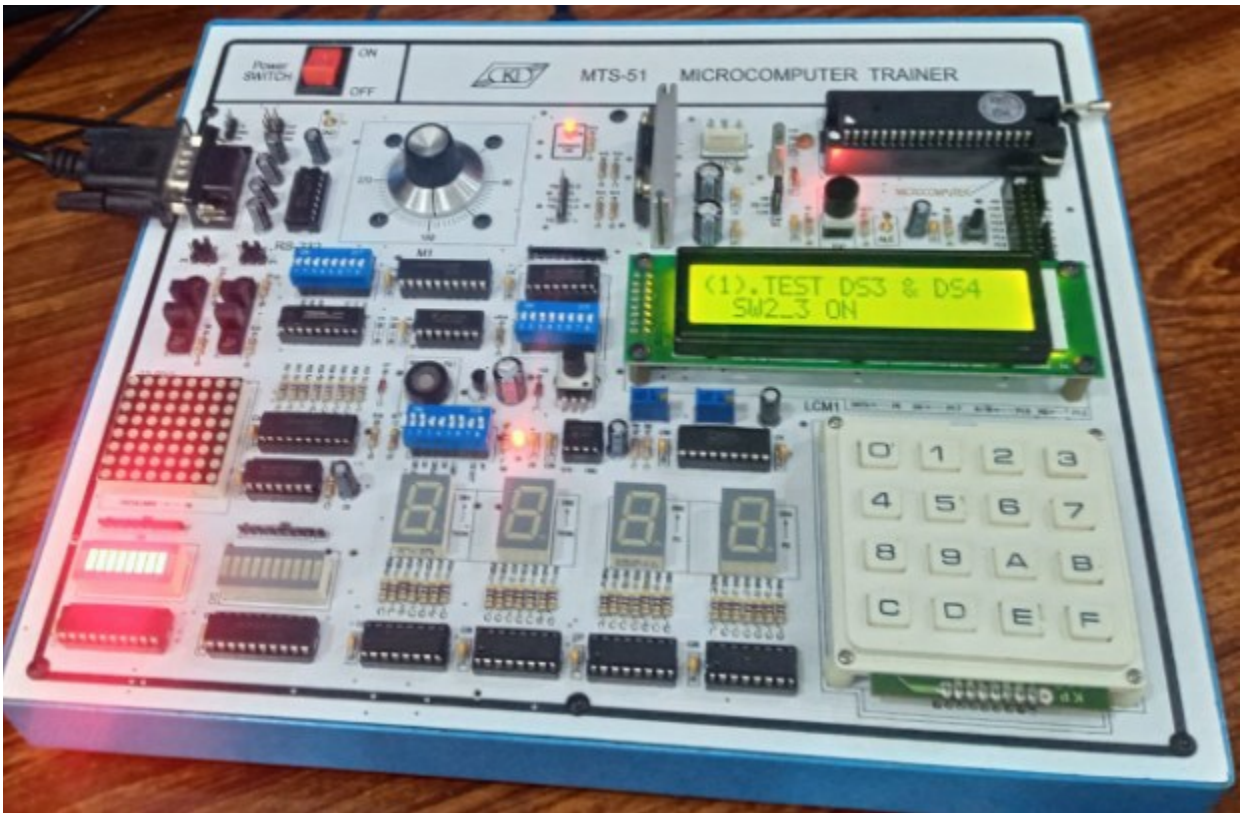
Experiment # 02

Introduction to MTS-51 Trainer board Inside's with demo examples

TASK #1:

Show the seven-segment display 3 and 4 by pressing 1. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

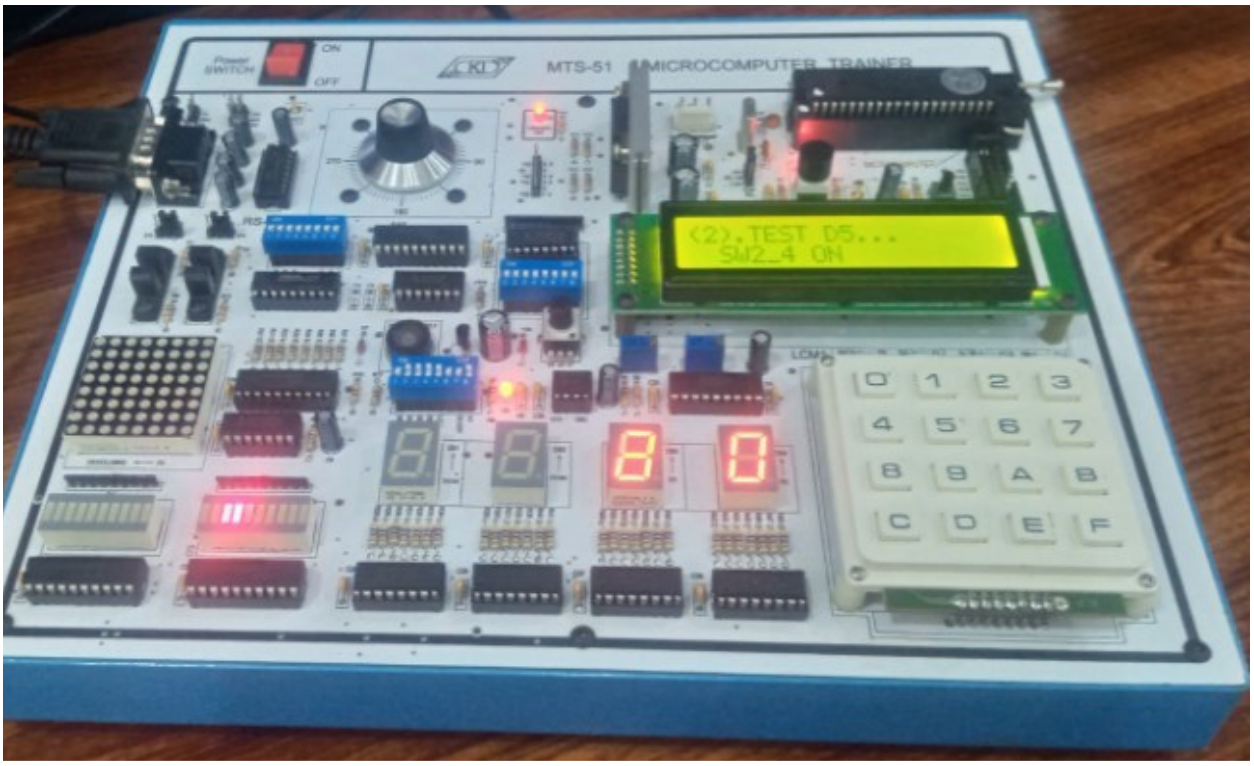
OUTPUT:



TASK #2:

Show the led bar 2 by pressing 2. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

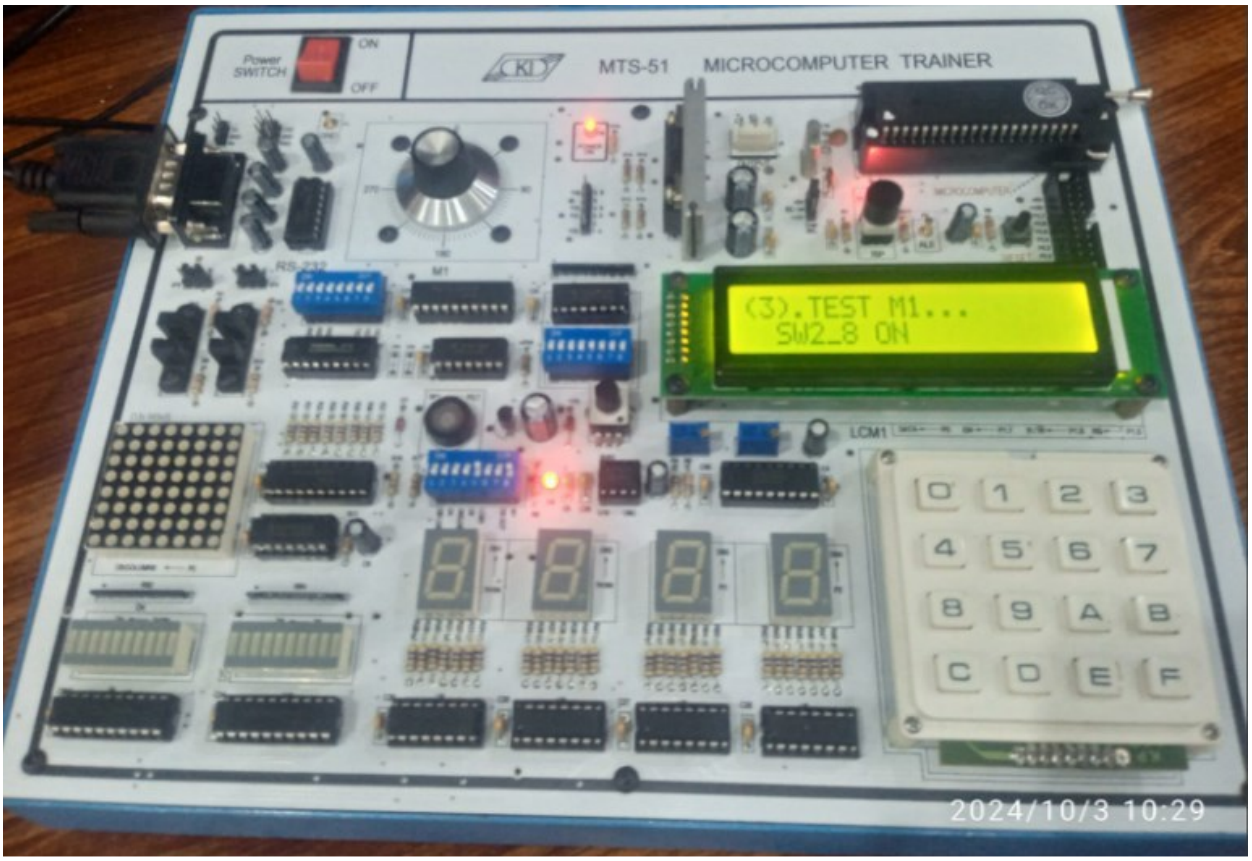
OUTPUT:



TASK #3:

Show the stepping motor by pressing 3. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

OUTPUT:



TASK #4:

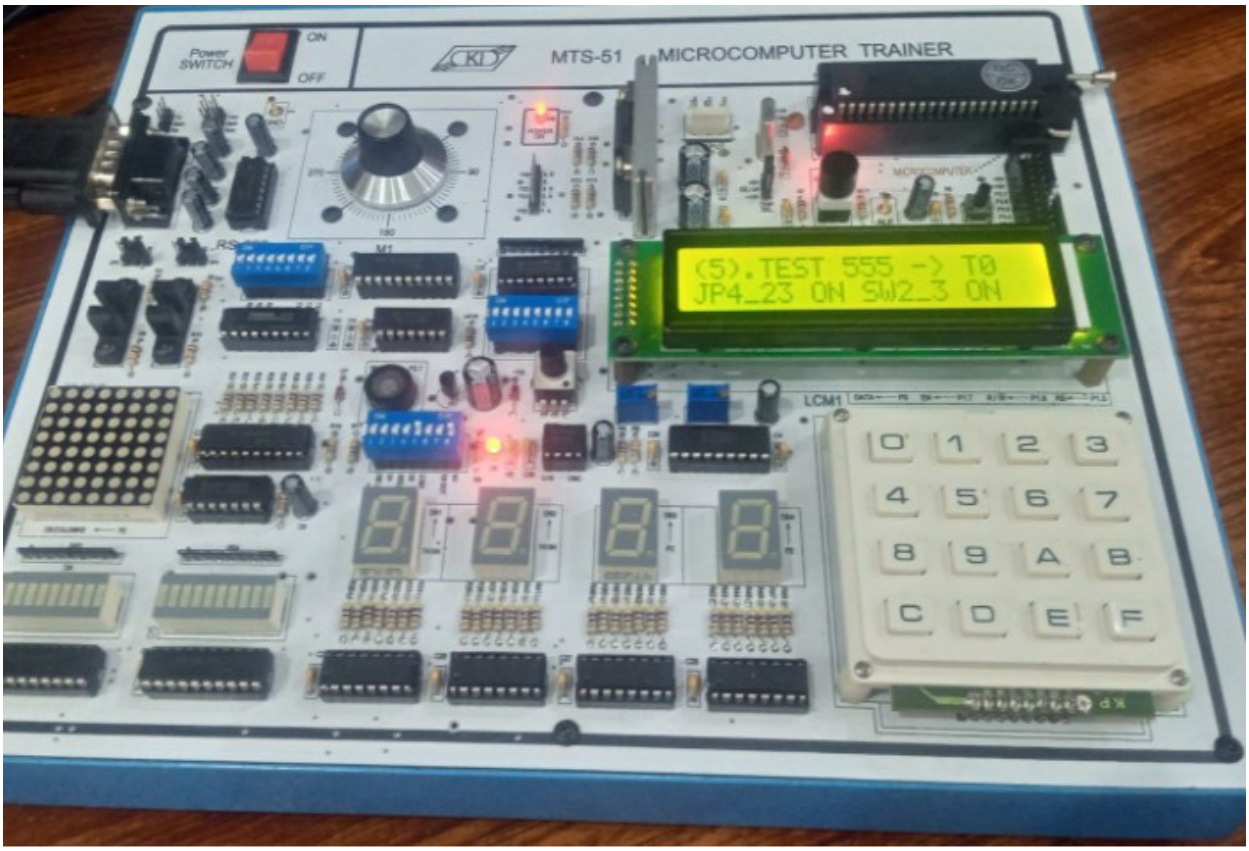
Show the IC 555 timer to speaker by pressing 4. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

OUTPUT:

TASK #5:

Show the IC 555 timer to seven segment display by pressing 5. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

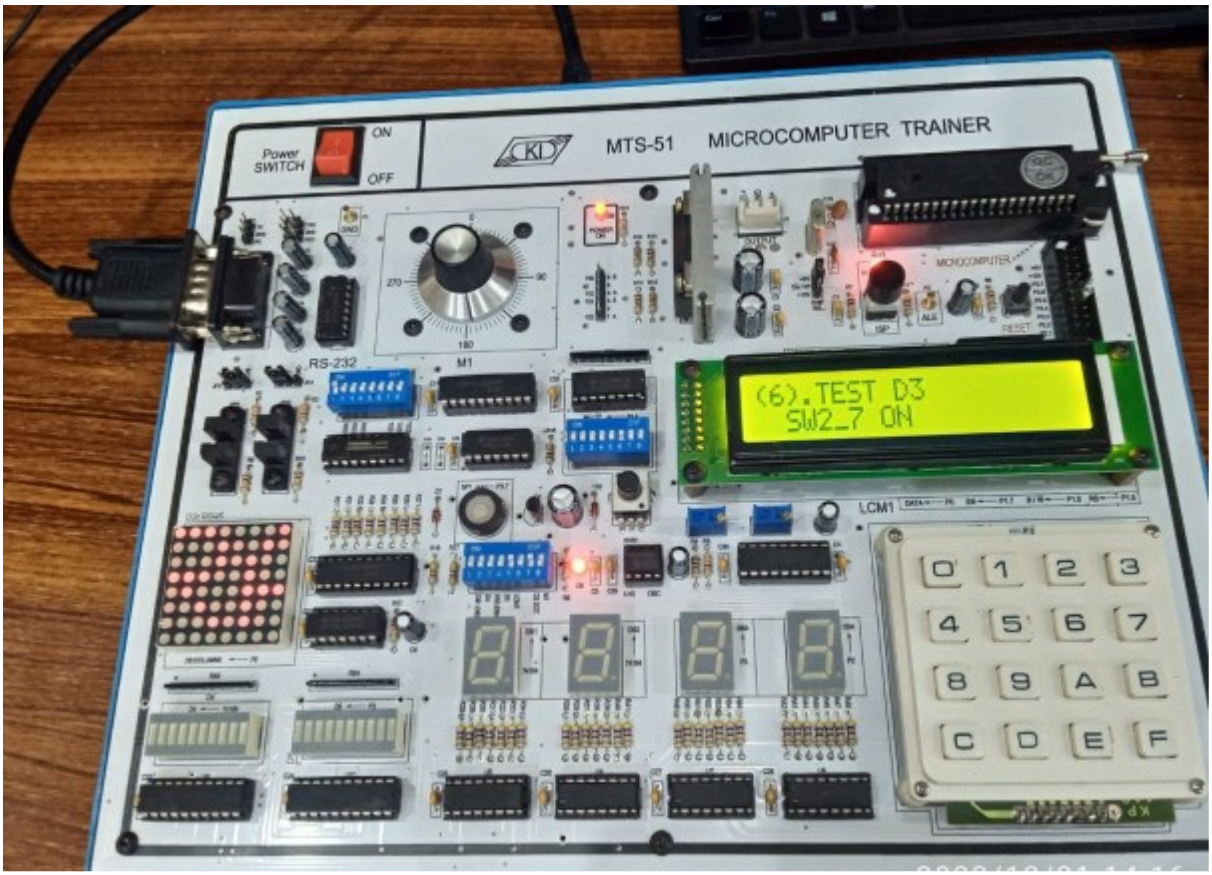
OUTPUT:



TASK #6:

Show the 8x8 dot matrix by pressing 6. Write down the steps to have a successful test.Take the snapshot of the final output generated by test.

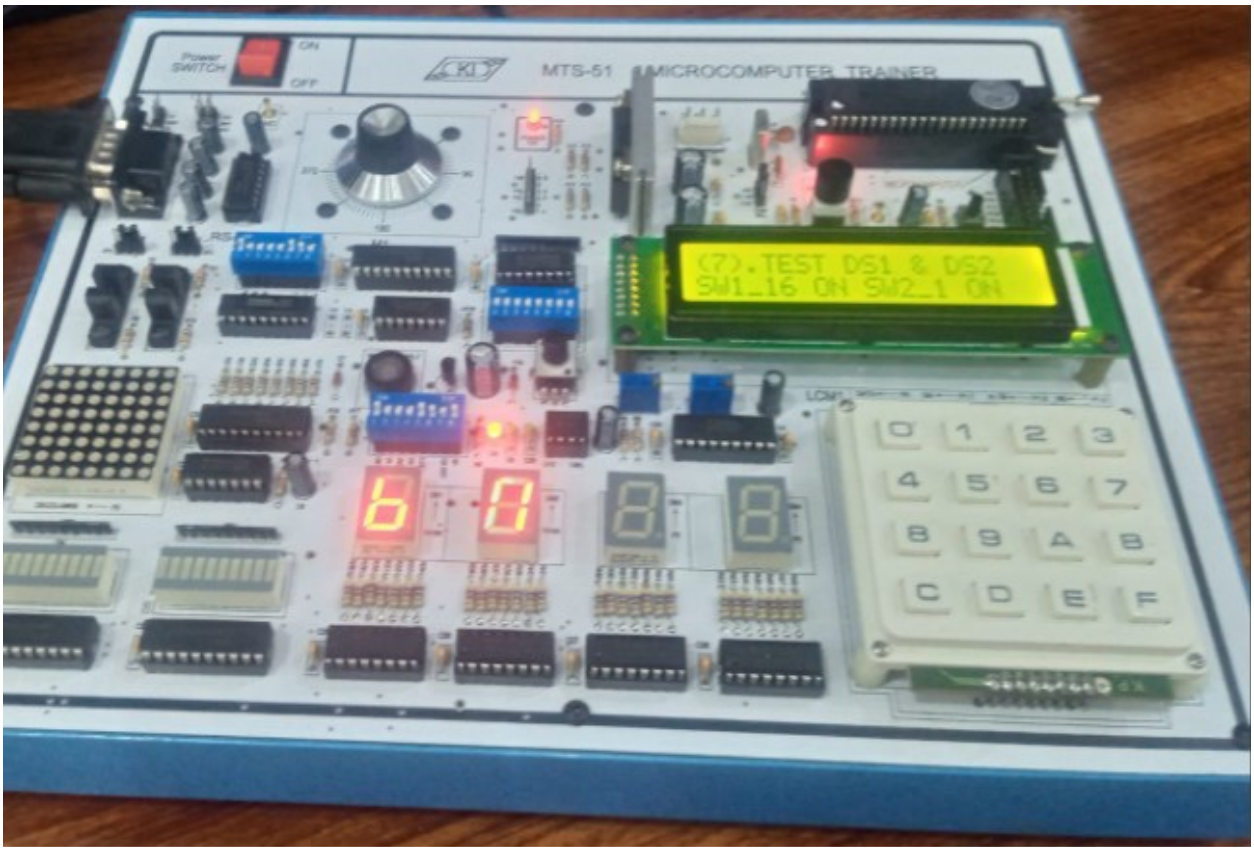
OUTPUT:



TASK #7:

Show the seven-segment display 1 and 2 by pressing 7. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

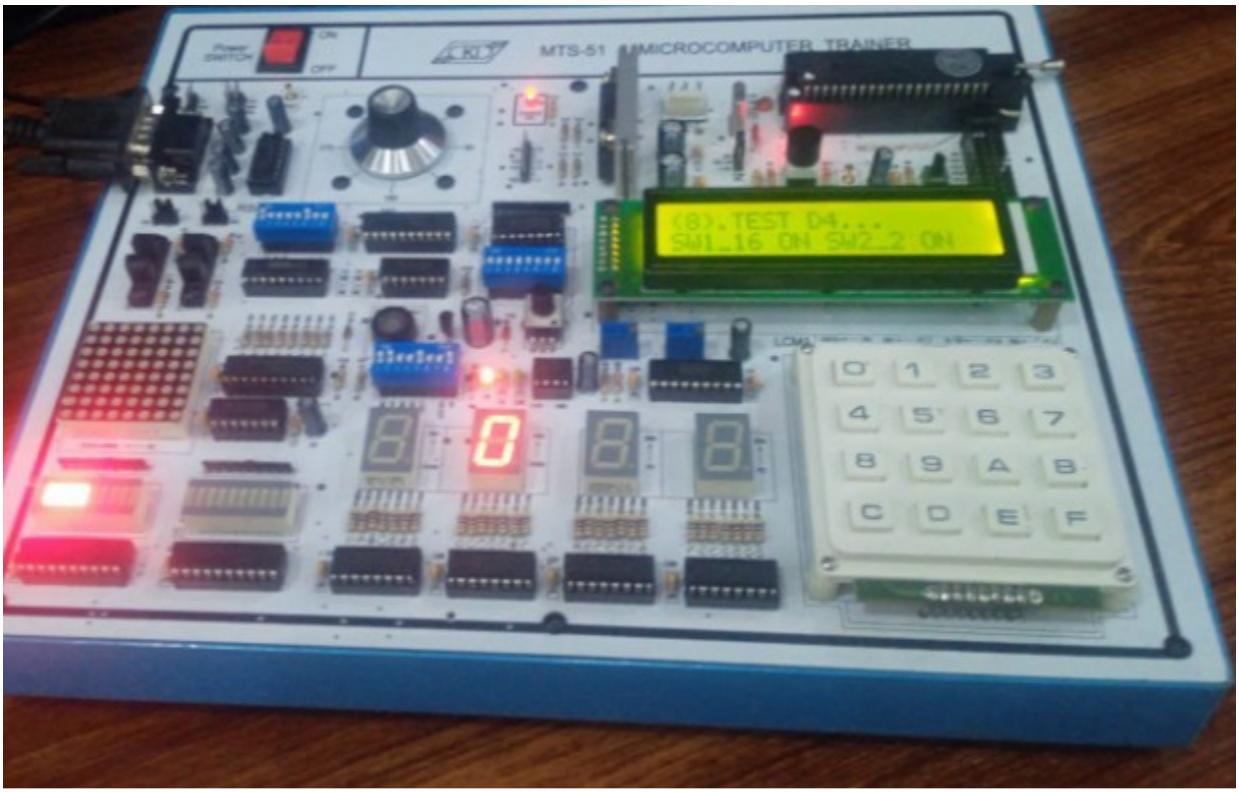
OUTPUT:



TASK #8:

Show the led bar 1 by pressing 8. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

OUTPUT:



TASK #9:

Show the photo interrupter (PH1) by pressing 9. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

OUTPUT:

TASK #10:

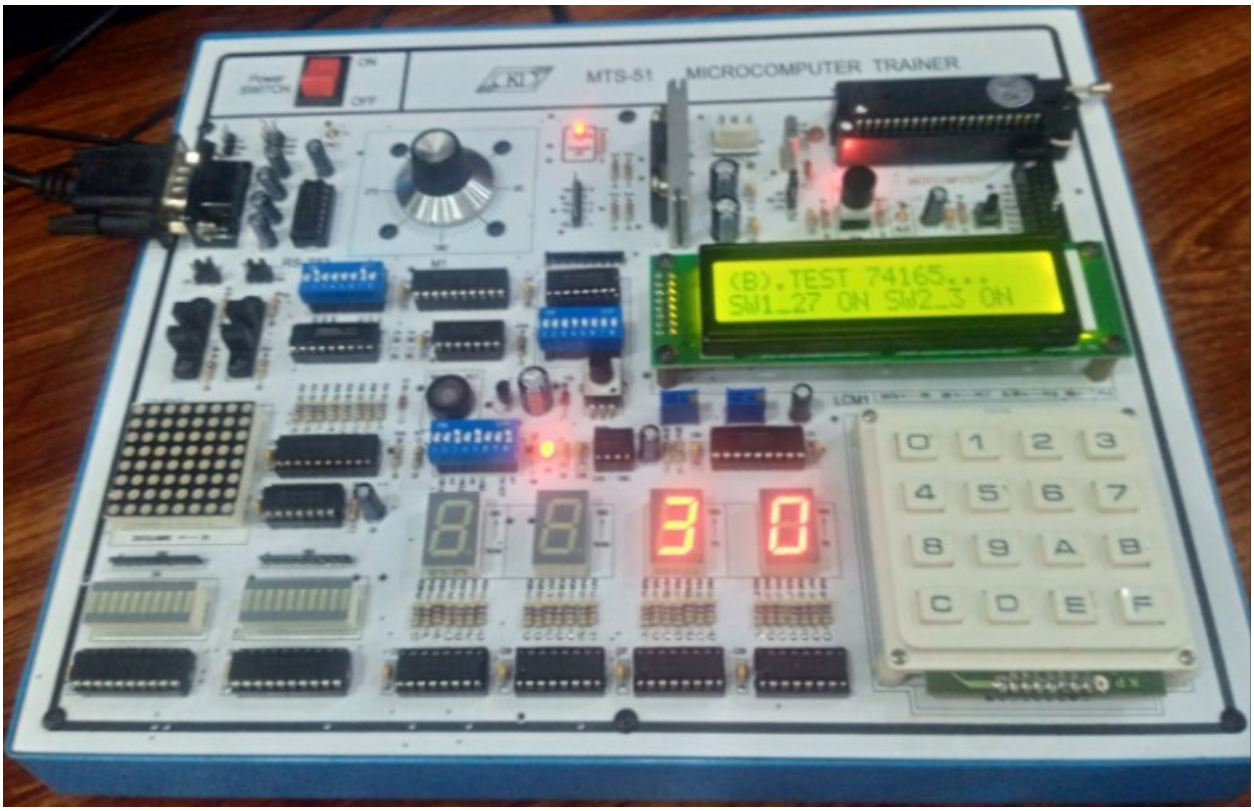
Show the photo interrupter (PH2) by pressing 10. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

OUTPUT:

TASK #11:

Show the parallel in serial out shift register by pressing 11. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

OUTPUT:



Experiment #3

Working of Flash Magic software with demo examples

TASK #1:

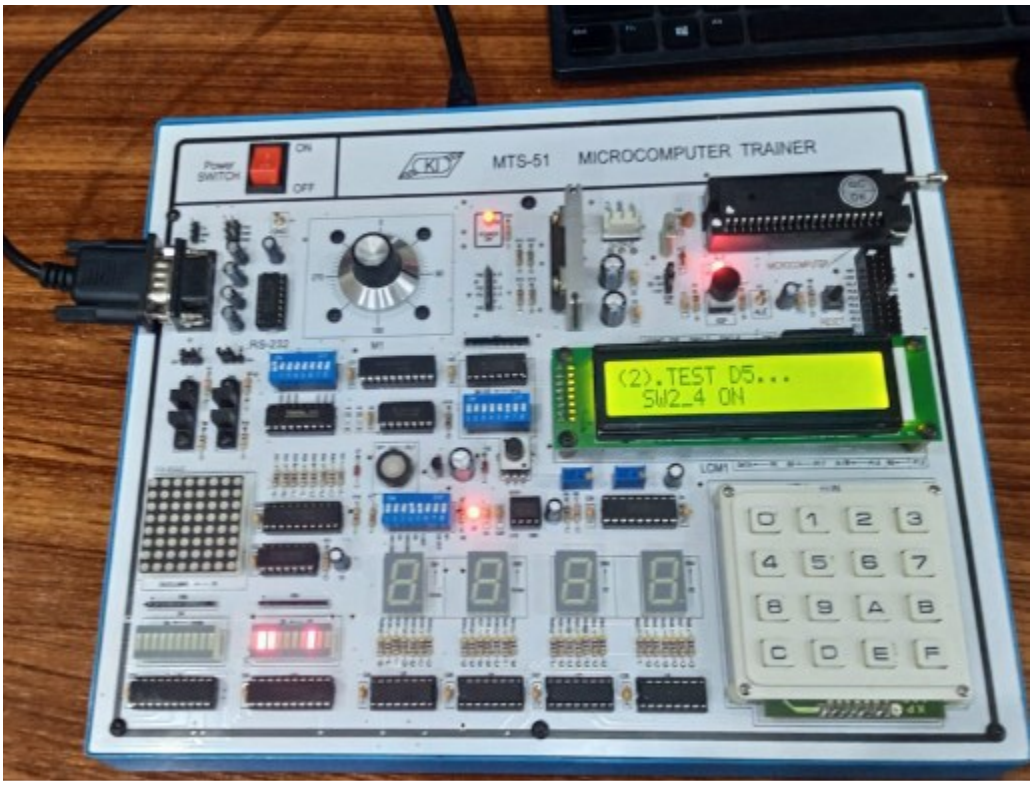
Demo example 1 for LED display control is demonstrated by configuring switches as in the previous lab. The steps for successful run are documented, and a snapshot of the final output is taken.

OUTPUT:

TASK #2:

Show the output of demo example 1 for 7 SEGMENT DISPLAY CONTROL (DS3 AND DS4) by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.

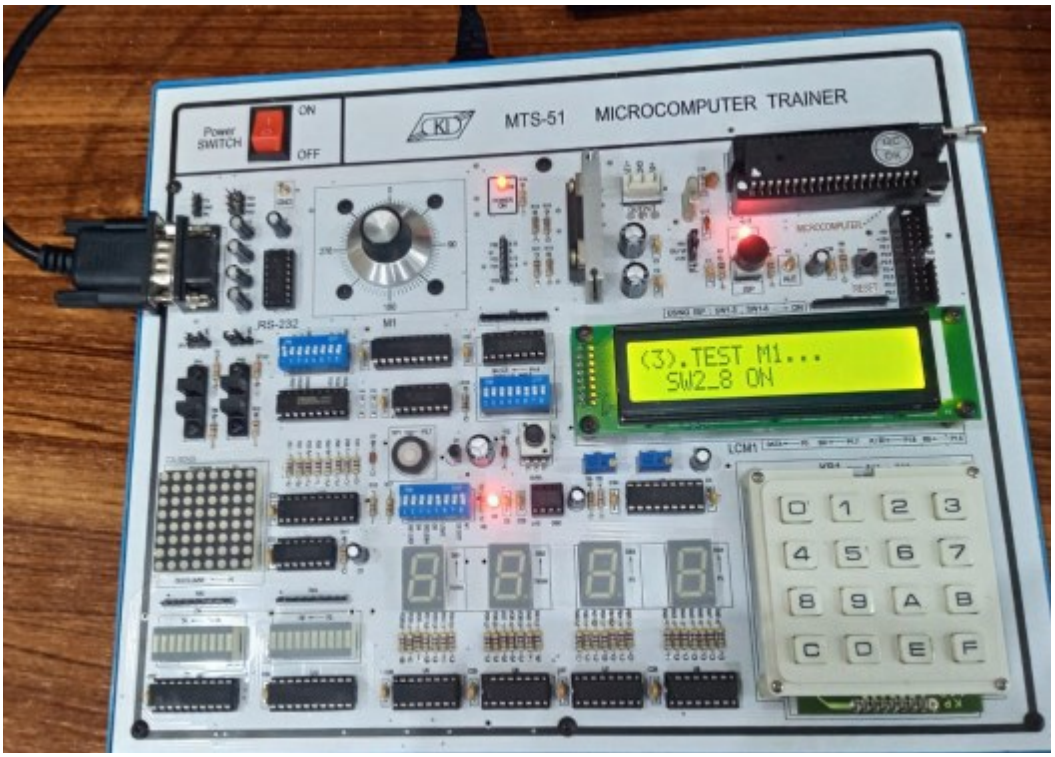
OUTPUT:



TASK #3:

Demo example 1 for DOT MATRIX LED CONTROL is demonstrated by configuring switches as in the previous lab, and the steps for a successful run are documented, along with a snapshot of the final output.

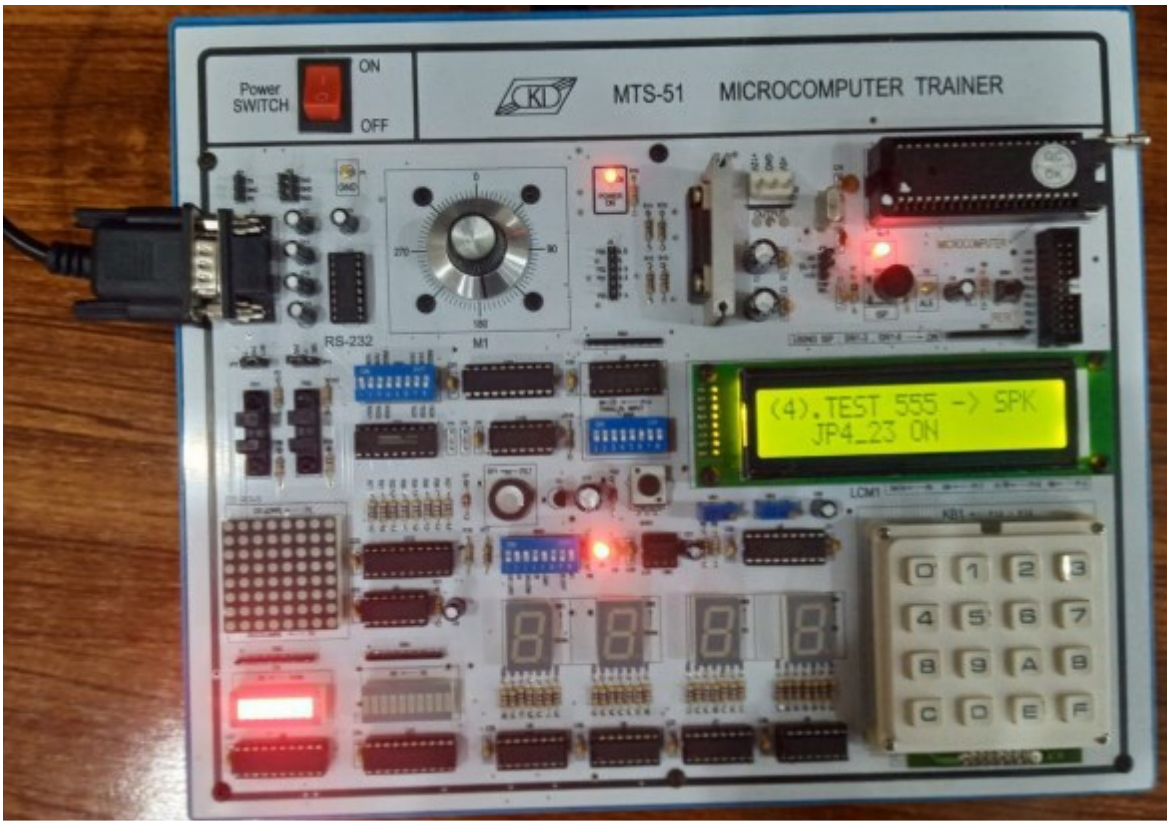
OUTPUT:



TASK #4:

Demo example 1 for MATRIX KEYBOARD CONTROL on DS4 is demonstrated by configuring switches as in the previous lab, and the steps for successful run are documented, along with a snapshot of the final output.

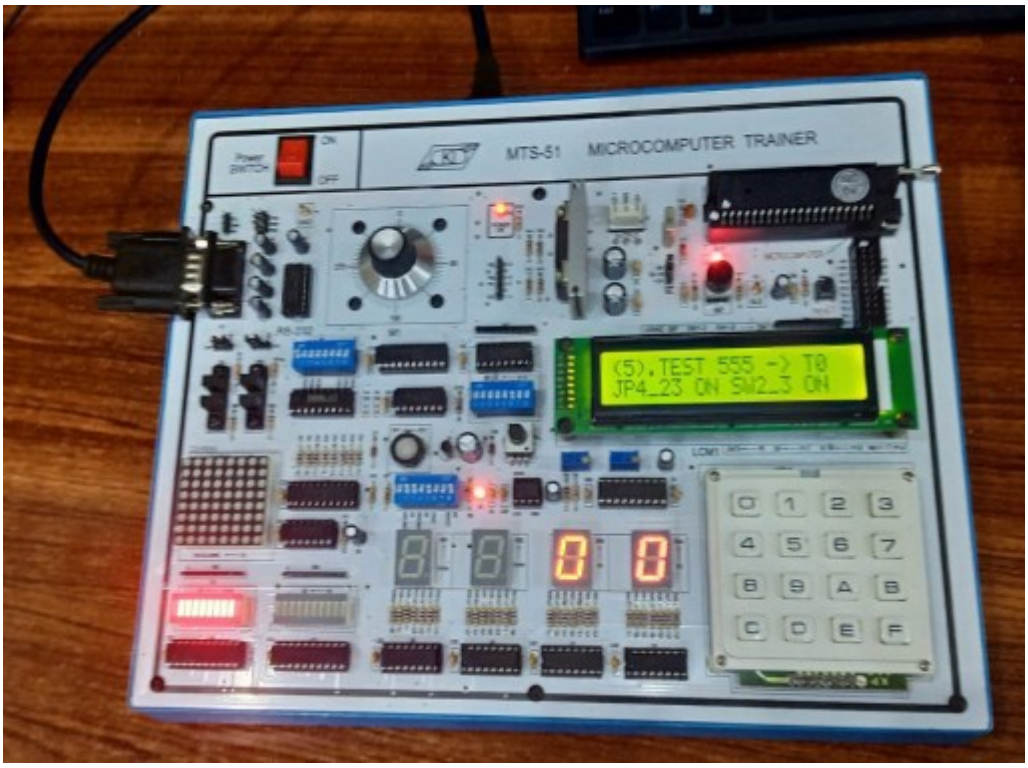
OUTPUT:



TASK #5:

Demo example 1 for STEP MOTOR CONTROL is demonstrated by configuring switches as in the previous lab, with steps for successful run recorded and a snapshot of the final output taken.

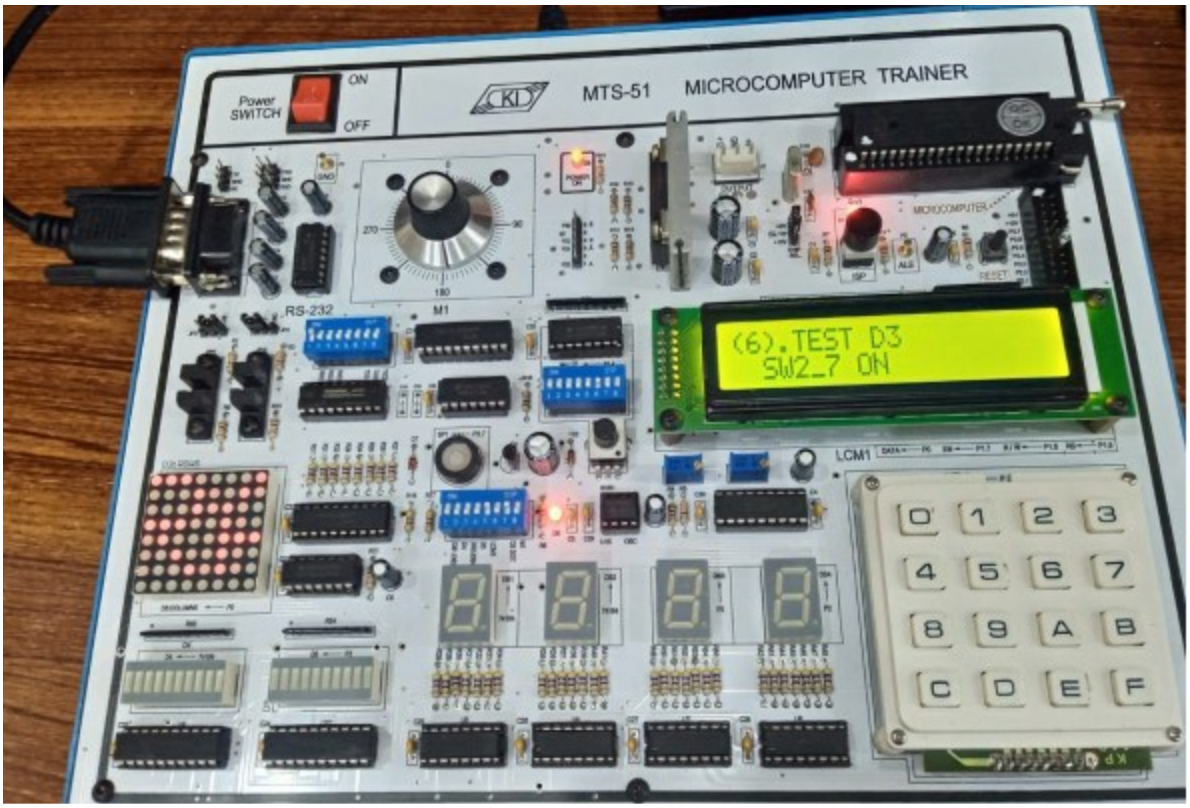
OUTPUT:



TASK #6:

Demo example 1 for PHOTO INTERRUPTER CONTROL (JP4=PH2 >T0) on STEP MOTOR is demonstrated, followed by configuring switches as in the previous lab, and a snapshot of the final output generated.

OUTPUT:



TASK #7:

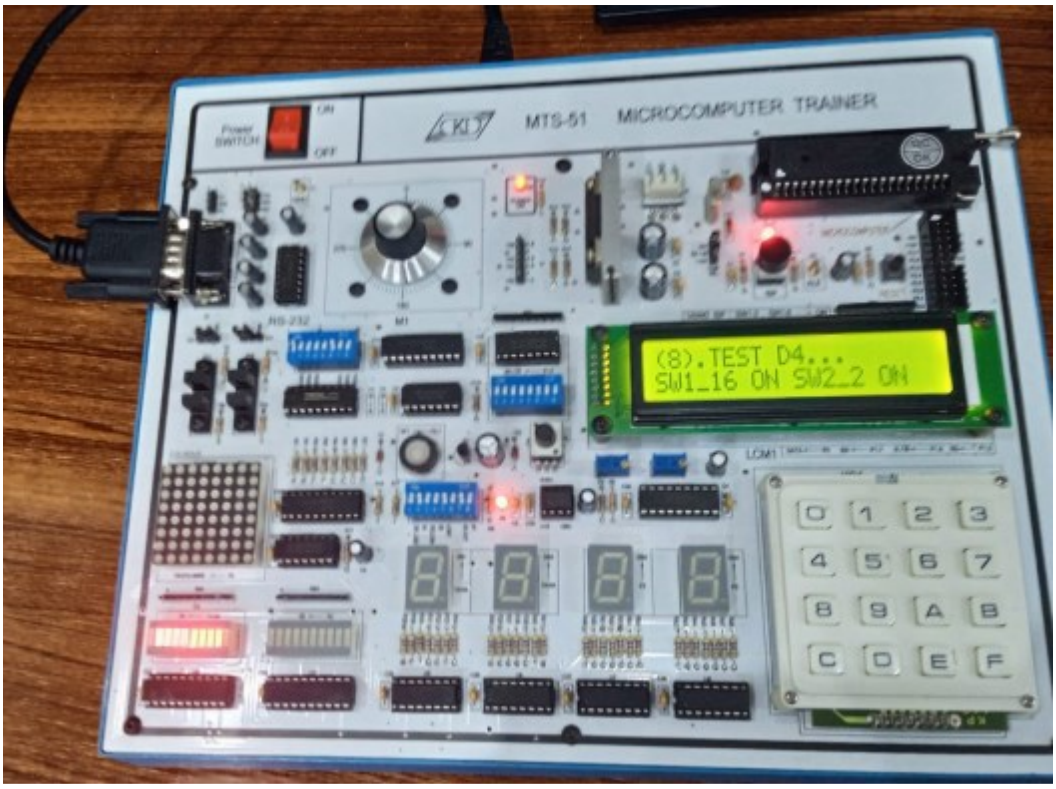
Demo example 1 for Pulse Counter (JP4=T0->OSC) on DS3 and DS4 is demonstrated, with switches configured as in the previous lab. A snapshot of the final output is taken.

OUTPUT:

TASK #8:

Demo example 1 for TIMER/COUNTER CONTROL on LED BAR is demonstrated by configuring switches as in the previous lab, with steps for successful run and a snapshot of the final output.

OUTPUT:



TASK #9:

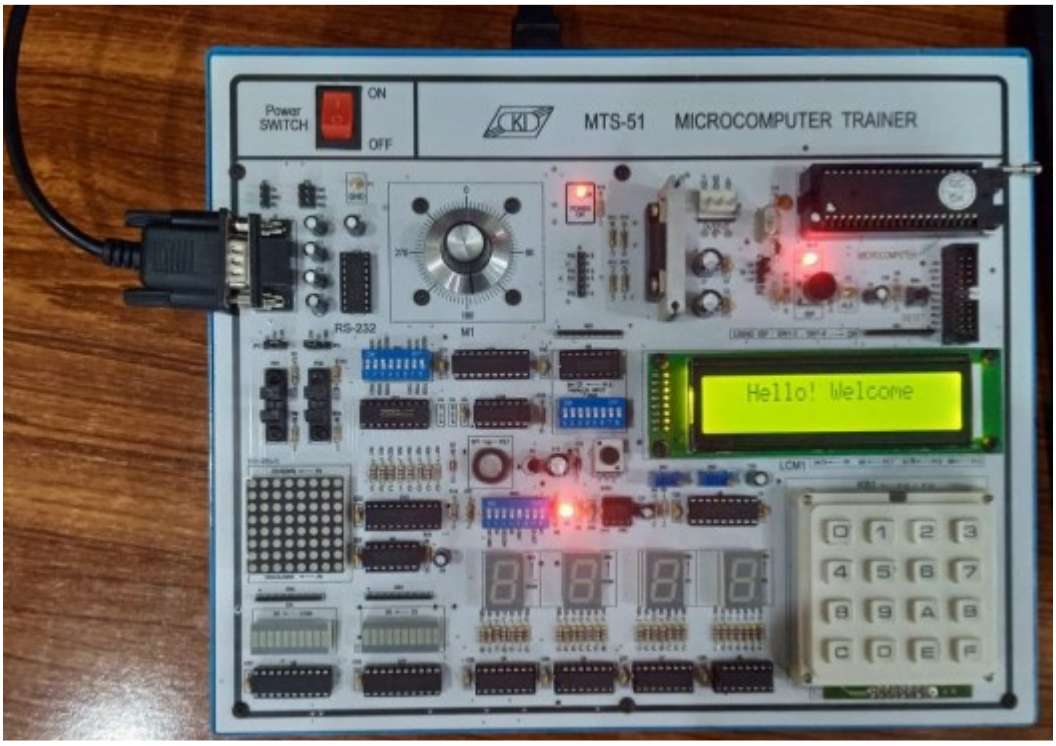
Demo example 1 for SPEAKER CONTROL output is displayed by configuring switches as in the previous lab, with steps for successful run documented and a snapshot taken of the final output.

OUTPUT:

TASK #10:

The task involves displaying the output of demo example 1 for LCM DISPLAY CONTROL, configuring switches as in the previous lab, recording the successful run steps, and taking a snapshot of the final output.

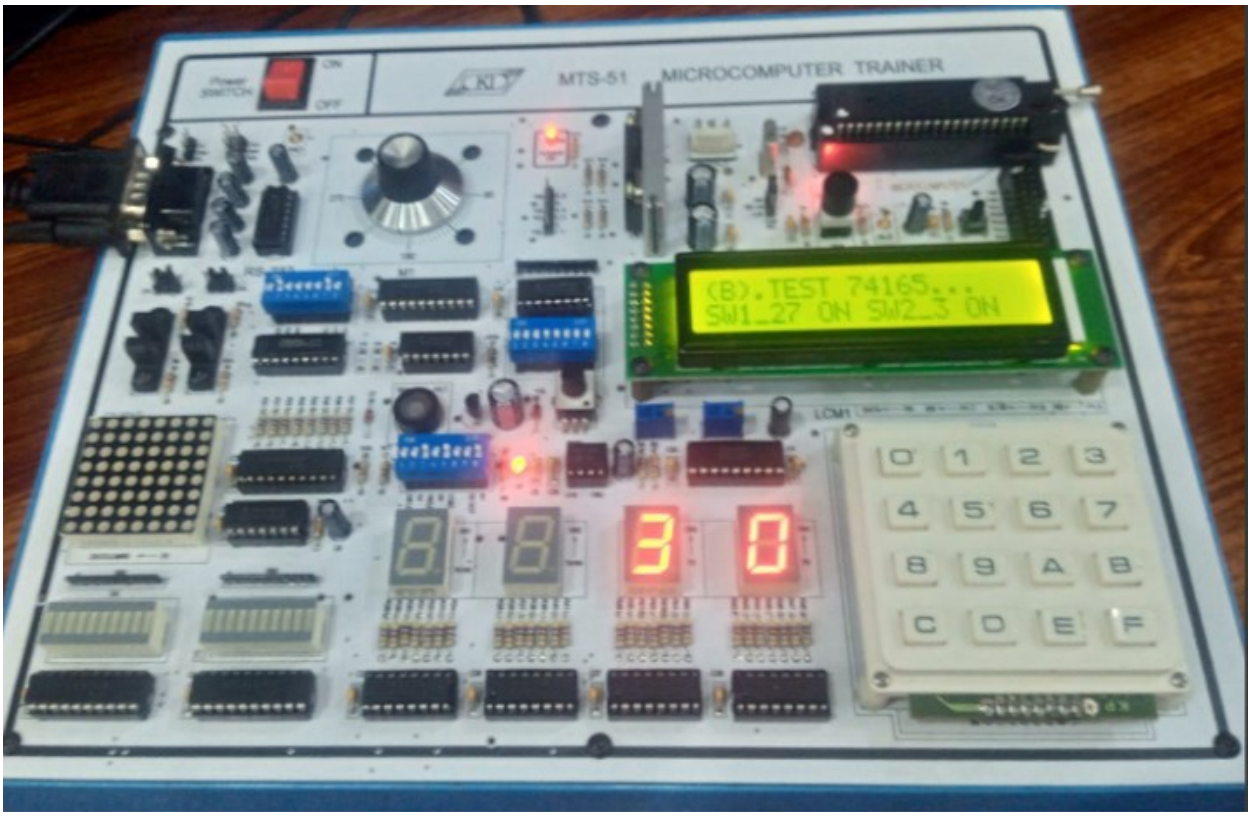
OUTPUT:



TASK#11:

Demo example 1 for SERIAL COMMUNICATION on DS4 of 2ND trainees is demonstrated by configuring switches as in the previous lab. The steps for successful run are documented, along with a snapshot of the final output.

OUTPUT:



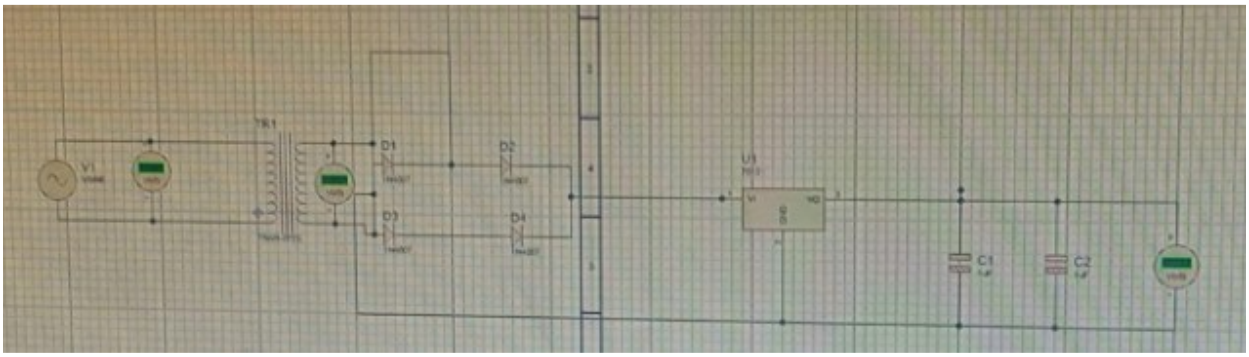
Experiment #4

Introduction to Proteus software with examples.

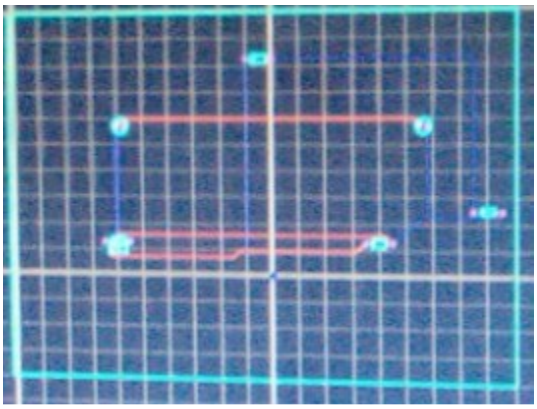
TASK#1:

In this task I had to make schematic diagram by using the proteus software and also convert the schematic diagram into pcb diagram.

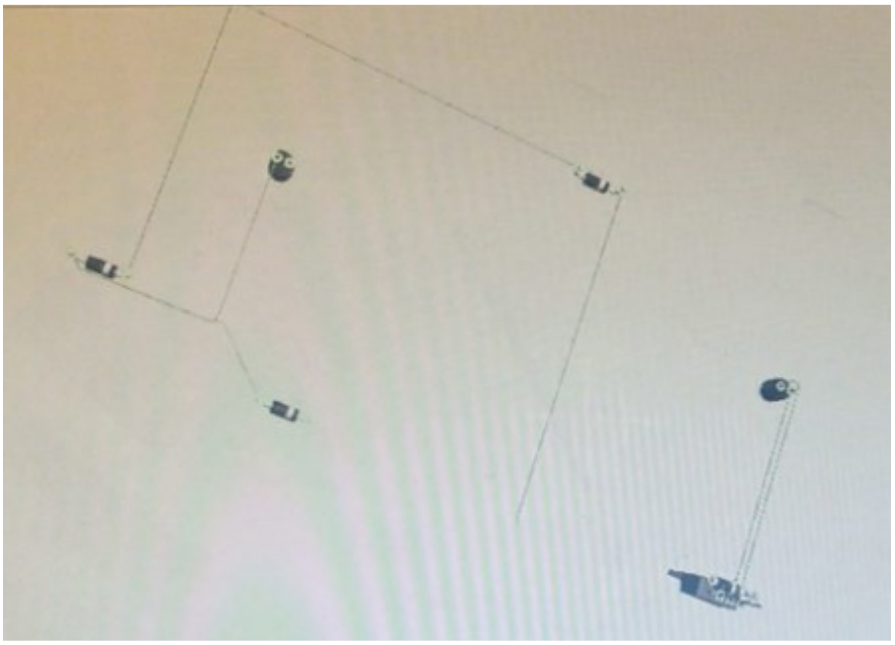
PROTEUS DIAGRAM:



PCB DIAGRAM:



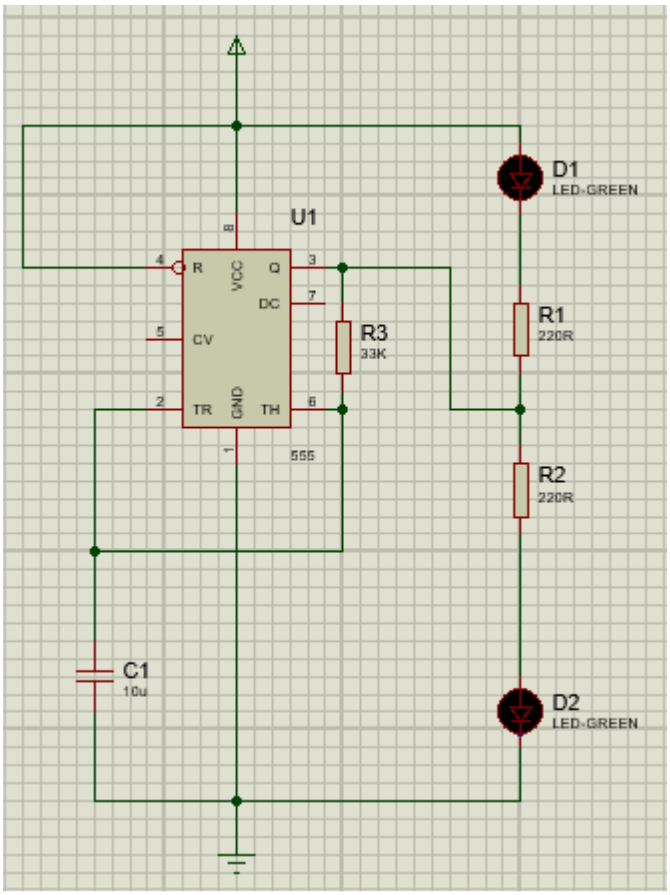
OUTPUT:



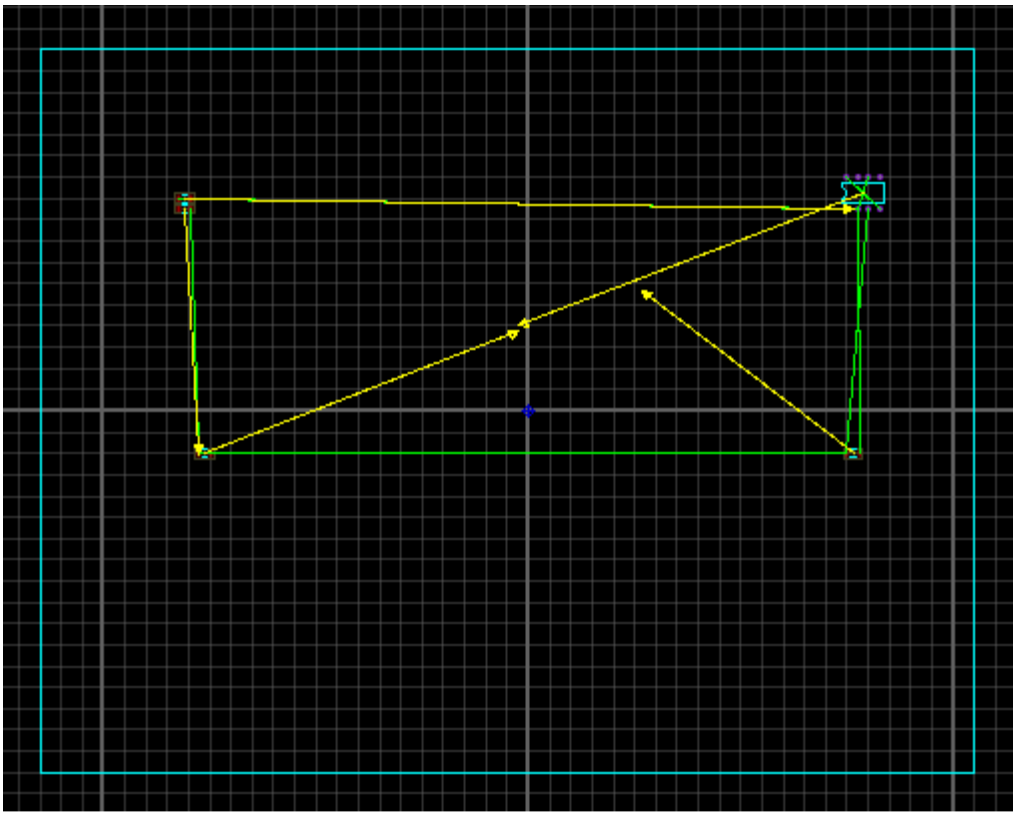
TASK#2:

In this task I had to make schematic diagram by using the proteus software and also convert the schematic diagram into pcb diagram. Show the output by flashing two LEDs with 555 Timer in Proteus ISIS by designing a circuit as well PCB layout shown in below figure.

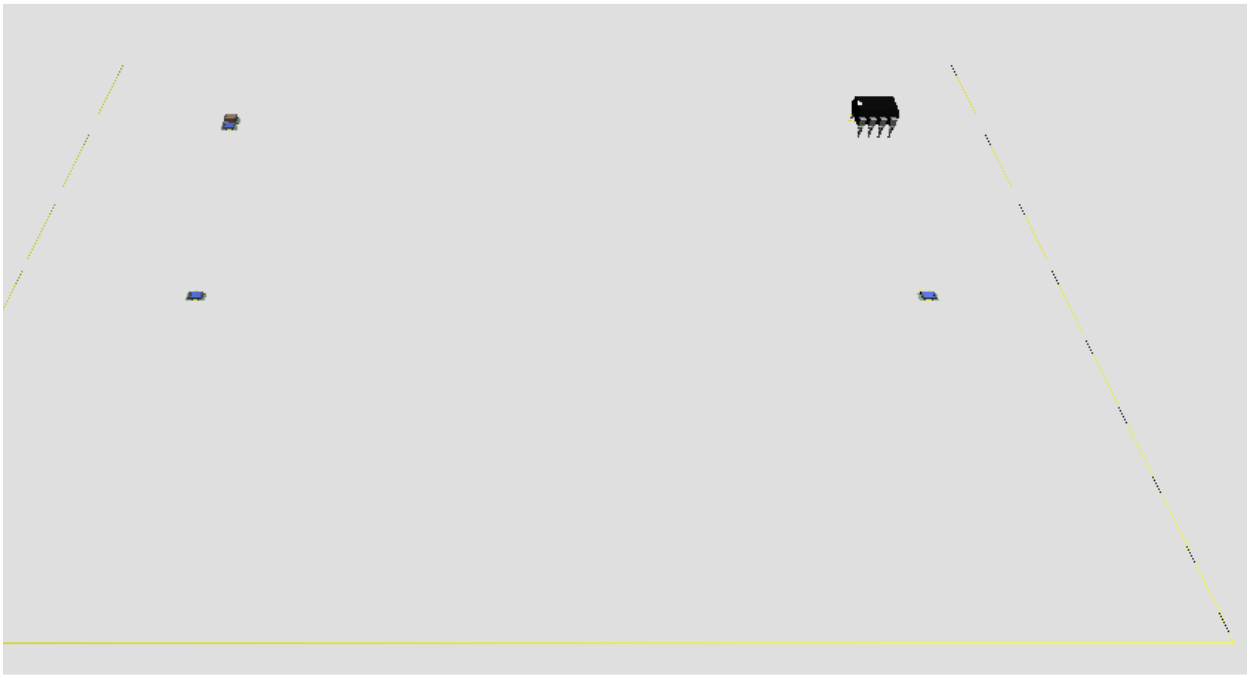
PROTEUS DIAGRAM:


D1
LED-GREEN
U1
8
4
R
VCC
Q
3
DC
7
5
CV
R3
33K
R1
220R
2
TR
GND
TH
6
1
555
R2
220R
C1
10u
D2
LED-GREEN

PCB DIAGRAM:



OUTPUT:



Experiment #5

Led Pattern generation using EdSim51 Simulator

TASK#1:

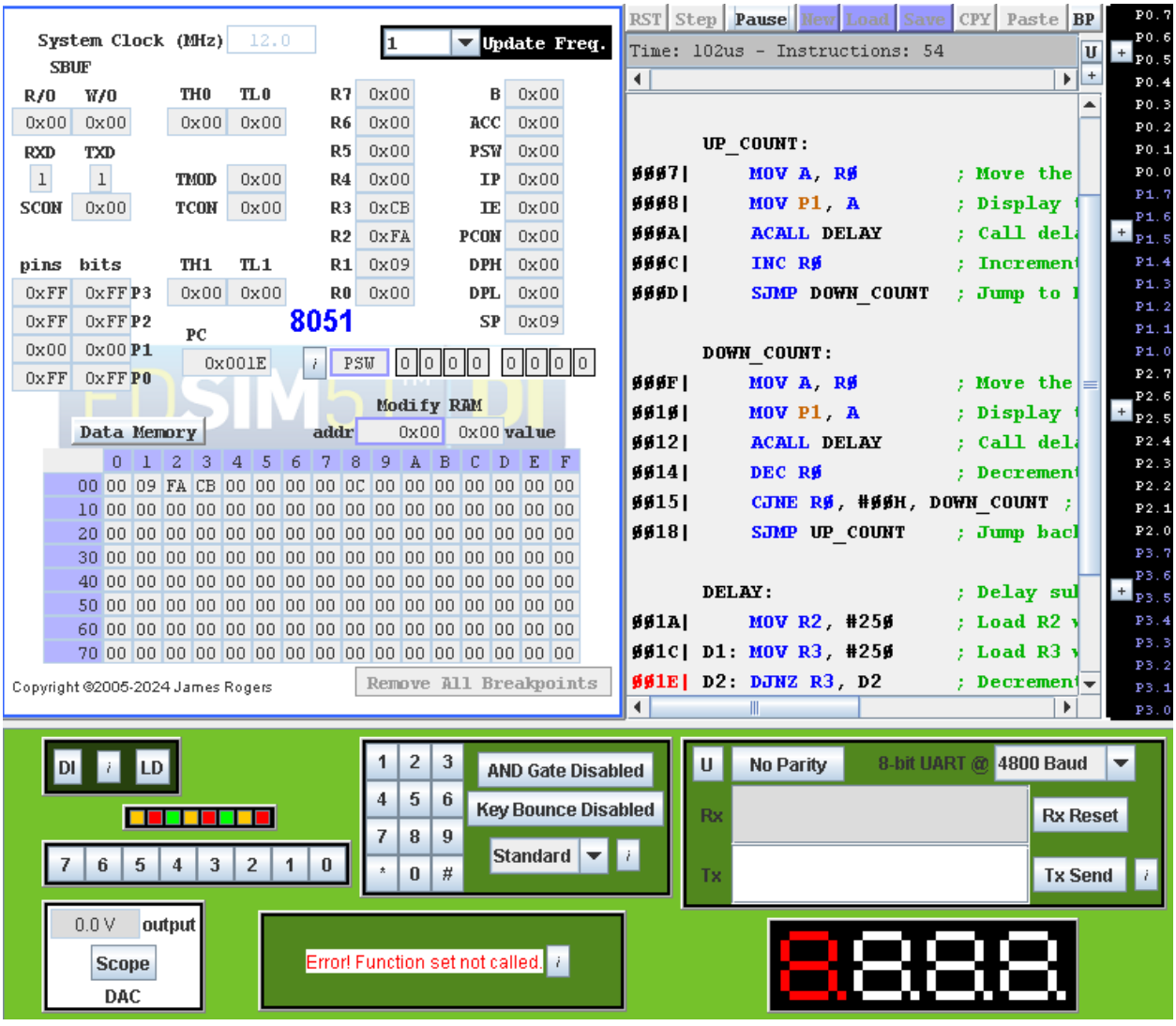
Write a program which act as up and down counter from 0 to 9 and 9 to 0 using the 8-LEDs.

CODE:

```
ORG 0000H           ; Start at address 0
MOV P1, #00H        ; Initialize Port 1 to 0 (all LEDs off)
MOV R0, #00H        ; Initialize counter value to 0
MOV R1, #09H        ; Set maximum count to 9
UP_COUNT:
   MOV A, R0        ; Move the counter value to the accumulator
   MOV P1, A        ; Display the counter value on LEDs
   ACALL DELAY      ; Call delay subroutine
   INC R0           ; Increment counter
   CJNE R0, R1, UP_COUNT  ; If R0 != 9, continue counting up
   SJMP DOWN_COUNT  ; Jump to DOWN_COUNT when reaching 9
DOWN_COUNT:
   MOV A, R0        ; Move the counter value to the accumulator
   MOV P1, A        ; Display the counter value on LEDs
   ACALL DELAY      ; Call delay subroutine
   DEC R0           ; Decrement counter
   CJNE R0, #00H, DOWN_COUNT ; If R0 != 0, continue counting down
   SJMP UP_COUNT    ; Jump back to UP_COUNT when reaching 0
DELAY:              ; Delay subroutine for LED visibility
```

```
    MOV R2, #250     ; Load R2 with 250 (outer loop counter)
D1: MOV R3, #250     ; Load R3 with 250 (inner loop counter)
D2: DJNZ R3, D2      ; Decrement R3 until it reaches 0
    DJNZ R2, D1      ; Decrement R2 until it reaches 0
    RET              ; Return from delay
END                  ; End of program
```

OUTPUT:



TASK#2:

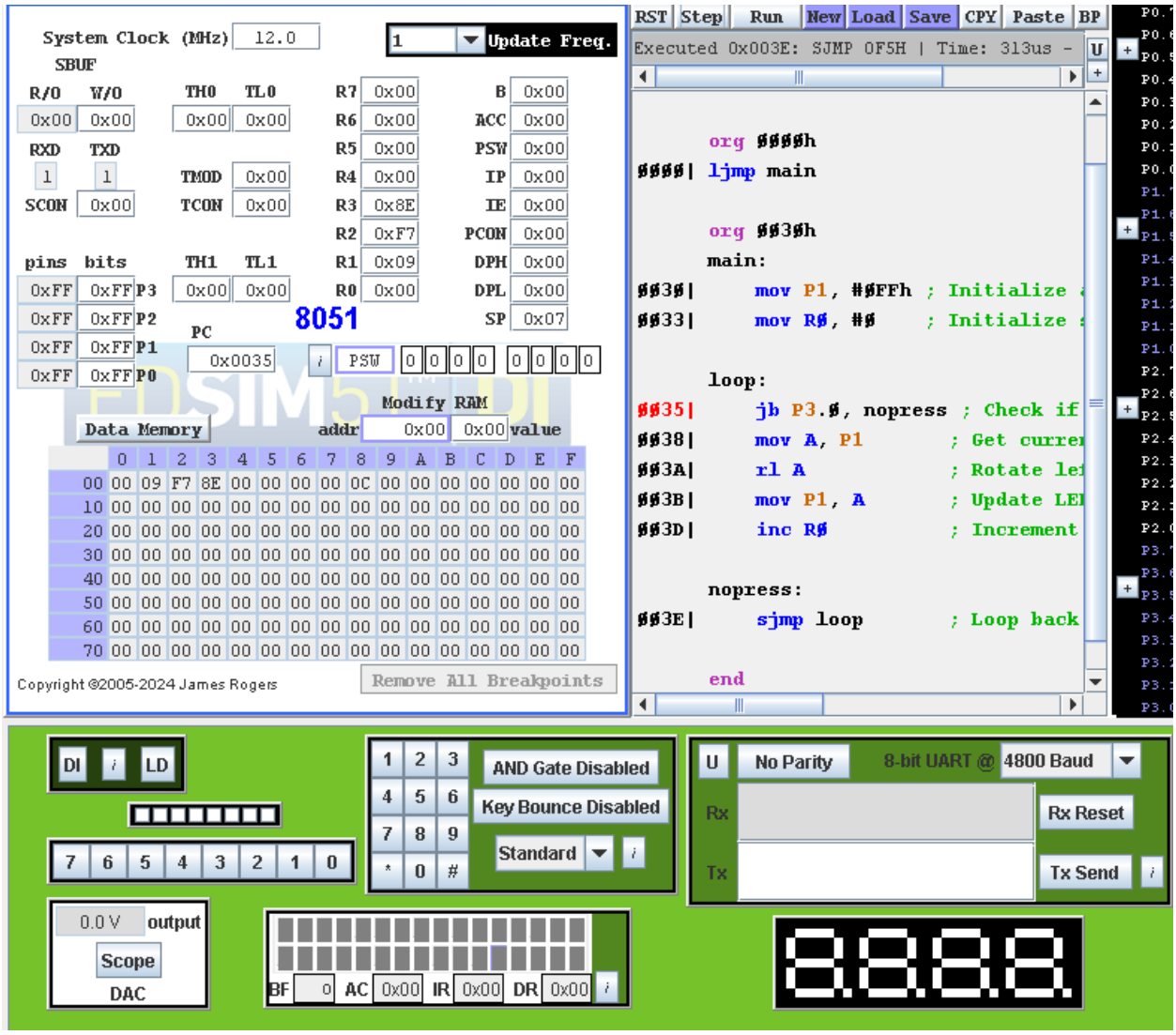
Write a program which turn on the rightmost led by pressing the leftmost switch and so on.

CODE:

```
ORG 0000H        ; Start of program
MOV P1, #0FFH    ; Set all LEDs off initially (P1 as output port, active low LEDs)
MOV P3, #0FFH    ; Set all switches off initially (P3 as input port)
MAIN:
   MOV A, P3     ; Read the switches on Port 3
   CPL A         ; Complement the bits to match active-low logic for switches
   MOV P1, #0FFH ; Reset LEDs to all OFF (high)
   JNB ACC.0, TURN_ON_LED7 ; If SW0 is pressed, turn on LED7 (rightmost)
   JNB ACC.1, TURN_ON_LED6 ; If SW1 is pressed, turn on LED6
   JNB ACC.2, TURN_ON_LED5 ; If SW2 is pressed, turn on LED5
   JNB ACC.3, TURN_ON_LED4 ; If SW3 is pressed, turn on LED4
   JNB ACC.4, TURN_ON_LED3 ; If SW4 is pressed, turn on LED3
   JNB ACC.5, TURN_ON_LED2 ; If SW5 is pressed, turn on LED2
   JNB ACC.6, TURN_ON_LED1 ; If SW6 is pressed, turn on LED1
   JNB ACC.7, TURN_ON_LED0 ; If SW7 is pressed, turn on LED0 (leftmost)
   SJMP MAIN     ; Repeat the loop
TURN_ON_LED7:
   MOV P1, #7FH ; 0111 1111, turns on LED7
```

```
    SJMP MAIN
TURN_ON_LED6:
    MOV P1, #BFH ; 1011 1111, turns on LED6
    SJMP MAIN
TURN_ON_LED5:
    MOV P1, #DFH ; 1101 1111, turns on LED5
    SJMP MAIN
TURN_ON_LED4:
    MOV P1, #EFH ; 1110 1111, turns on LED4
    SJMP MAIN
TURN_ON_LED3:
    MOV P1, #F7H ; 1111 0111, turns on LED3
    SJMP MAIN
TURN_ON_LED2:
    MOV P1, #FBH ; 1111 1011, turns on LED2
    SJMP MAIN
TURN_ON_LED1:
    MOV P1, #FDH ; 1111 1101, turns on LED1
    SJMP MAIN
TURN_ON_LED0:
    MOV P1, #FEH ; 1111 1110, turns on LED0
    SJMP MAIN
END
```

OUTPUT:



TASK#3:

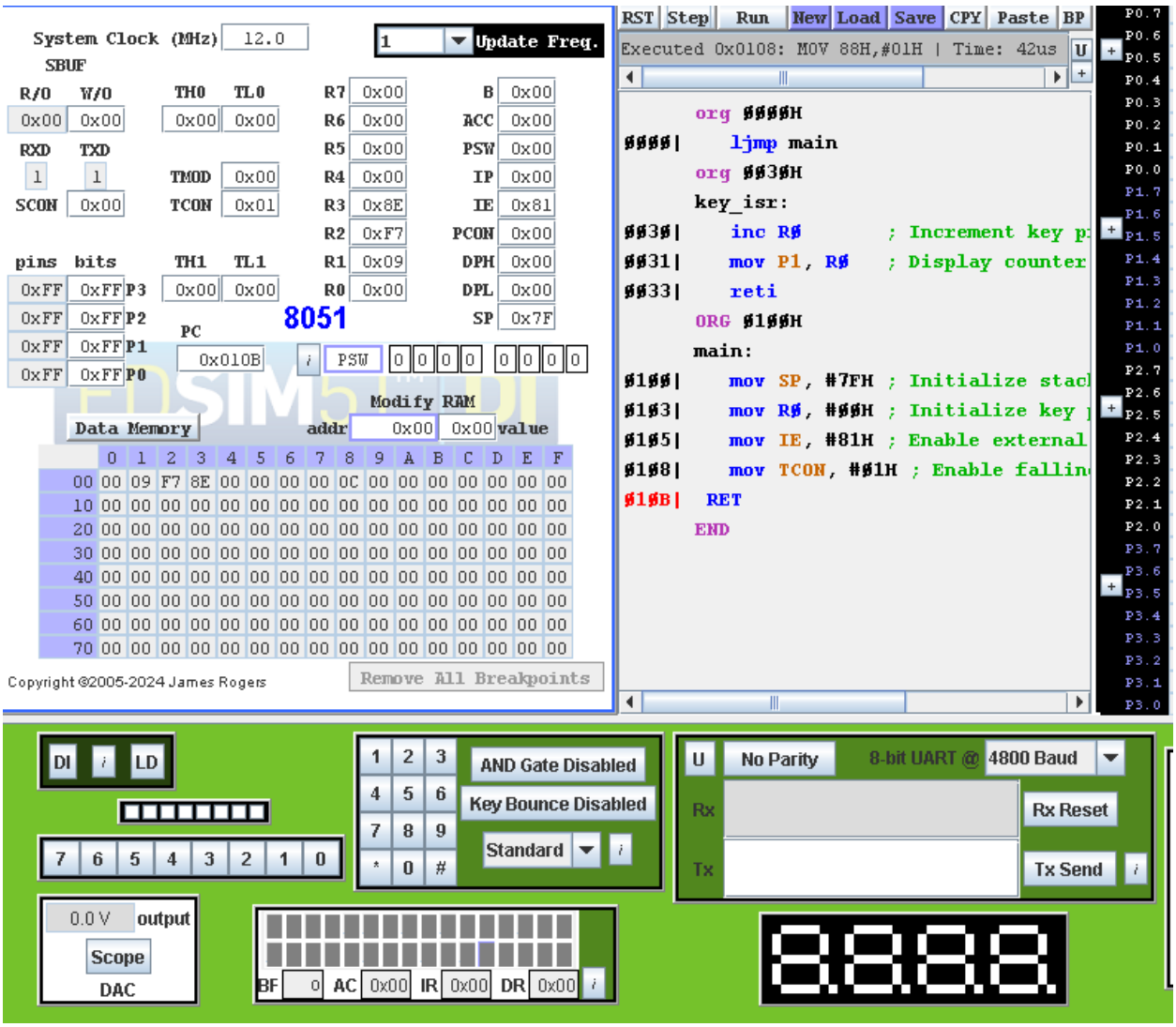
Write a program code which shows the number of key pressed on LED bar.

CODE:

```
$NOLIST
$MODLP51
$LIST
```

```
org 0000H
  ljmp main
org 0030H
key_isr:
  inc R0       ; Increment key press counter in R0
  mov P1, R0   ; Display counter value on LED bar
  reti
ORG 0100H
main:
  mov SP, #7FH ; Initialize stack pointer
  mov R0, #00H ; Initialize key press counter to 0
  mov IE, #81H ; Enable external interrupt 0 (P3.2)
  mov TCON, #01H ; Enable falling edge detection on P3.2
  ljmp $
END
```

OUTPUT:



Experiment #6

Introduction to Keil Debugger and simulator with Led Display Control using MTS-51.

TASK#1:

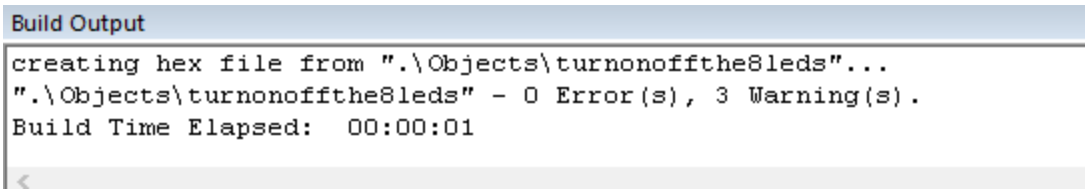
Write a program which Sequentially and repeatedly turn on/off the 8 LEDs from the leftmost LED to the rightmost LED. The ON time of each LED is 0.1 seconds.

CODE:

```
ORG 0000H        ; Start at address 0
```

```
MOV P1, #00H      ; Clear all LEDs initially (assuming LEDs are
connected to Port 1)
MOV R0, #08H      ; Counter for 8 LEDs


MAIN:
   MOV A, #01H   ; Initialize with leftmost LED (binary 00000001)
   MOV R1, #08H  ; Load the loop count for 8 LEDs
LOOP:
   MOV P1, A     ; Output current LED state to Port 1
   ACALL DELAY   ; Delay for 0.1 seconds
   RL A          ; Rotate left to shift the bit to the next LED
   DJNZ R1, LOOP ; Decrement R1, repeat for 8 LEDs
   SJMP MAIN     ; Jump back to start for continuous operation
DELAY:           ; 0.1-second delay subroutine (approximate)
   MOV R2, #250
DELAY_LOOP1:
   MOV R3, #250
DELAY_LOOP2:
   DJNZ R3, DELAY_LOOP2
   DJNZ R2, DELAY_LOOP1
   RET
END
```

OUTPUT:



TASK#2:

Write a program which Sequentially turn on/off the 8 LEDs from the leftmost LED to the rightmost LED. When the rightmost LED is reached, the sequence is inverted. The ON time of each LED is 0.1 seconds.

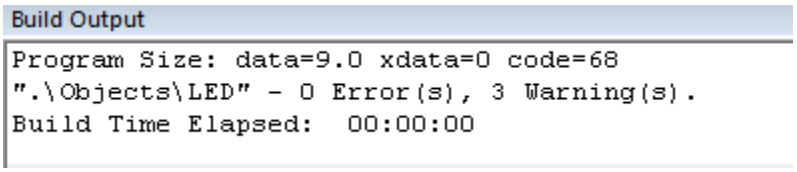
CODE:

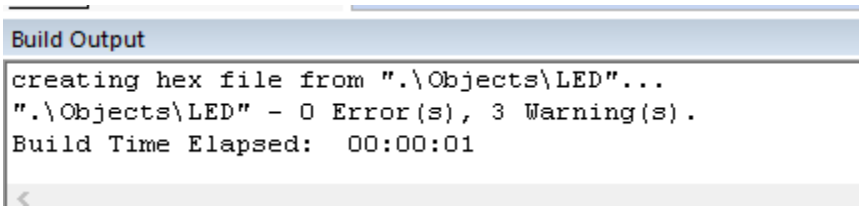
```
ORG 0000H          ; Start of code memory
START:
    MOV P1, #01H   ; Initialize the leftmost LED (bit 0 of Port 1)
    MOV R0, #01H   ; Set up the direction for the sequence (left to right)
LOOP:
    MOV P1, A      ; Update the Port 1 to the value of A register
    ACALL DELAY    ; Call the delay subroutine
    MOV A, P1      ; Load the current state of LEDs into A
    RLC A          ; Rotate left through carry (move to next LED in sequence)
    MOV P1, A      ; Update the Port 1 with the new value
    ACALL DELAY    ; Call the delay subroutine
    JNB P1.7, LOOP ; Jump back to LOOP if the rightmost LED is not reached
    ; Invert the direction (right to left)
    MOV P1, #80H   ; Set the rightmost LED
```

```
    MOV R0, #80H   ; Set up the direction for the sequence (right to left)
INVERT_LOOP:
    MOV P1, A      ; Update Port 1 with the current value
    ACALL DELAY    ; Call the delay subroutine
    MOV A, P1      ; Load the current state of LEDs into A
    RRC A          ; Rotate right through carry (move to previous LED in sequence)
    MOV P1, A      ; Update the Port 1 with the new value
    ACALL DELAY    ; Call the delay subroutine
    JNB P1.0, INVERT_LOOP ; Jump back to INVERT_LOOP if the leftmost LED is not reached
    ; Repeat the sequence
    SJMP LOOP      ; Jump back to the start of the loop
DELAY:
    MOV R1, #200   ; Outer loop count
DELAY1:
    MOV R2, #250   ; Inner loop count
DELAY2:
    NOP            ; No Operation (used for delay)
    NOP
    NOP
    NOP
    DJNZ R2, DELAY2 ; Decrement inner loop counter
    DJNZ R1, DELAY1 ; Decrement outer loop counter
```

```
    RET             ; Return from the delay subroutine
END
```

OUTPUT:





TASK#3:

Write a program which Sequentially turn on/off the two LEDs from two ends towards the middle. When the middle reached, the sequence is inverted. The ON time of each LED is 1 seconds.

CODE:

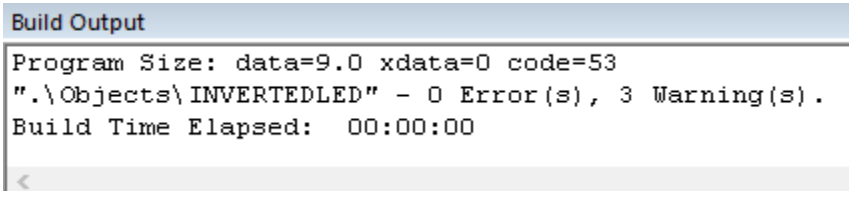
```
ORG 0000H           ; Start of program memory
START:
    MOV P1, #0x01       ; Turn on the leftmost LED (P1.0)
    ACALL DELAY         ; Wait for 1 second
    MOV P1, #0x00       ; Turn off the leftmost LED (P1.0)
    ACALL DELAY         ; Wait for brief time before next action
    MOV P1, #0x80       ; Turn on the rightmost LED (P1.7)
    ACALL DELAY         ; Wait for 1 second
    MOV P1, #0x00       ; Turn off the rightmost LED (P1.7)
    ACALL DELAY         ; Wait for brief time before next action
```

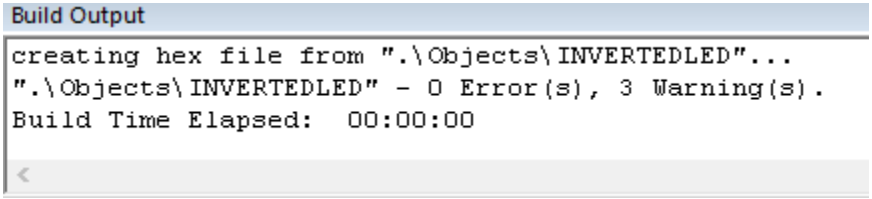
```
    MOV R0, #0          ; Initialize R0 as a counter for middle
    MOV R1, #7          ; Set R1 to 7 (for P1.7)
    ; Move towards the middle from both ends
LOOP:
    MOV A, P1           ; Check the current status of LEDs
    INC R0              ; Increment counter from left side (P1.0)
    MOV P1, #0x01 << R0  ; Turn on the next LED from left end
    ACALL DELAY         ; Wait for 1 second
    DEC R1              ; Decrement counter from right side (P1.7)
    MOV P1, #0x80 >> R1  ; Turn on the next LED from the right end
    ACALL DELAY         ; Wait for 1 second
    ; Check if the middle is reached
    CJNE R0, R1, LOOP    ; If the left end and right end don't meet, continue
    SJMP INVERT_SEQUENCE ; If the middle is reached, invert sequence
; Invert the sequence and go back to the end
INVERT_SEQUENCE:
    MOV R0, #7          ; Set R0 back to the rightmost end (P1.7)
    MOV R1, #0          ; Set R1 back to the leftmost end (P1.0)
REVERSE_LOOP:
    MOV A, P1           ; Check current LED state
    DEC R0              ; Move towards left from right end
    MOV P1, #0x01 << R0 ; Turn on next LED from right to left
```

```
    ACALL DELAY         ; Wait for 1 second
    INC R1              ; Move towards right from left end
    MOV P1, #0x80 >> R1 ; Turn on next LED from left to right
    ACALL DELAY         ; Wait for 1 second
    ; Check if the middle is reached again
    CJNE R0, R1, REVERSE_LOOP ; Continue until the middle is reached
    SJMP START          ; Repeat the cycle
; Delay subroutine (approx. 1 second)
DELAY:
    MOV R2, #250        ; Outer loop count
DELAY1:
    MOV R3, #250        ; Inner loop count
DELAY2:
    NOP                 ; No operation (just for delay)
    NOP
    NOP
    NOP
    NOP
    NOP
    NOP
    NOP
    NOP
```

```
    NOP
    NOP
    NOP
    NOP
    NOP
    NOP
    NOP
    NOP
    NOP
    DJNZ R3, DELAY2    ; Decrement and loop if not zero
    DJNZ R2, DELAY1    ; Decrement and loop if not zero
    RET
END
```

OUTPUT:





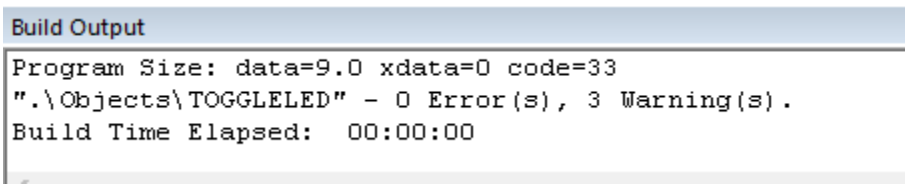
TASK#4:

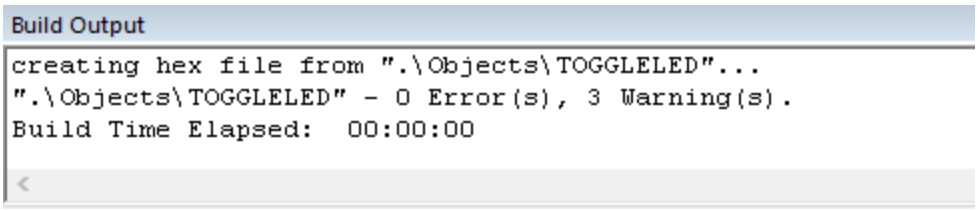
Write a program which toggle between the even led and odd led having the delay of 2 sec.

CODE:

```
; Program to toggle between even and odd LEDs on an 8051 microcontroller
; using Keil Assembly Language
ORG 0000H          ; Origin, start of the program
START:             ; Start label
   MOV P1, #0  ; Initialize Port 1 to 0 (all LEDs off)
MAIN_LOOP:         ; Main loop label
   MOV A, P1   ; Move current state of Port 1 to Accumulator A
   CPL A       ; Complement the accumulator (toggle the state)
   MOV P1, A   ; Output the toggled state back to Port 1
   ACALL DELAY ; Call delay subroutine
   SJMP MAIN_LOOP ; Jump back to main loop
DELAY:             ; Delay subroutine
   MOV R0, #20 ; Load R0 with a value for delay
DELAY_LOOP:
   NOP         ; No operation (do nothing)
   DJNZ R0, DELAY_LOOP ; Decrement R0 and repeat if not zero
   RET         ; Return from subroutine
END                ; End of program
```

OUTPUT:





TASK#5:

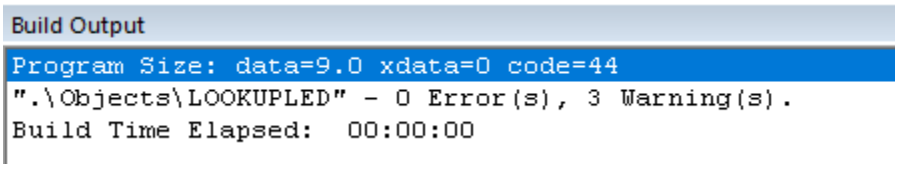
Write a program which Uses the technique of table lookup to show the LED patterns resided in memory. Each pattern stays on 0.1 seconds.

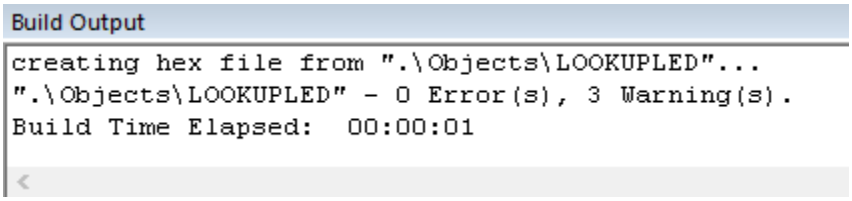
CODE:

```
; 8051 Assembly Code for LED Patterns using Table Lookup

; This program demonstrates LED patterns stored in memory.

; Each pattern stays on for 0.1 seconds.

ORG 0000H       ; Origin, start of program memory

; Define the LED patterns in a lookup table

PATTERNS: DB 0x01, 0x02, 0x04, 0x08, 0x10, 0x20, 0x40, 0x80 ; 8 patterns for LEDs

; Define constants

DELAY_COUNT EQU 250 ; Adjust this value to achieve approximately 0.1 seconds delay

; Main program starts here

START:

   MOV DPTR, #PATTERNS ; Point to the start of the patterns table

   MOV R0, #08         ; Set R0 to number of patterns (8)

MAIN_LOOP:

   CLR A               ; Clear accumulator
```

```
    MOVC A, @A+DPTR        ; Load pattern from lookup table into accumulator
    MOV P1, A              ; Output the pattern to Port 1 (LEDs connected here)
    ACALL DELAY            ; Call delay subroutine
    INC DPTR               ; Move to the next pattern in the table
    DJNZ R0, MAIN_LOOP     ; Decrement R0 and repeat if not zero
    SJMP START             ; Repeat the main loop indefinitely
DELAY:
    MOV R1, #DELAY_COUNT ; Load delay count into R1
DELAY_LOOP:
    DJNZ R1, DELAY_LOOP  ; Decrement R1 until it reaches zero
    RET                    ; Return from delay subroutine
END                        ; End of program
```

OUTPUT:





TASK#6:

Write a program which Turn on LEDs according to the following cases and repeat each

case 8 times.

a. Case 1: Turn on LEDs from left to right.

b. Case 1: Turn on LEDs from right to left.

c. Blink two sets (left and right sides) of LEDs alternately.

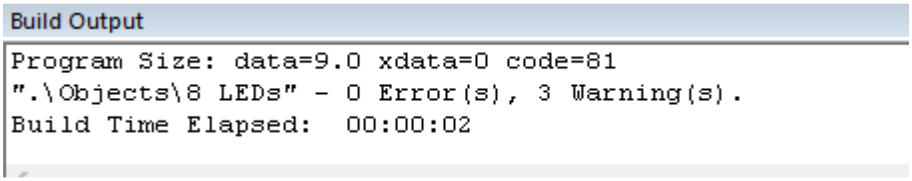
d. Blink all of 8 LEDs.

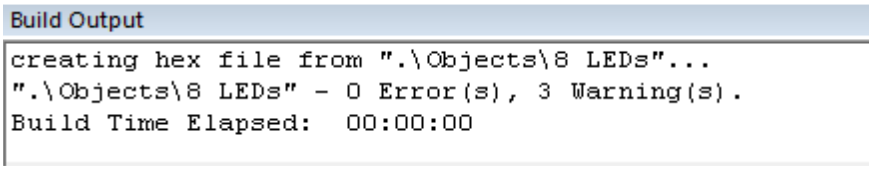
OUTPUT:

```
ORG 0000H
START:
MOV P1, #00H
MOV R0, #08H
CASE1_LEFT:
MOV R1, #00H
LEFT_LOOP:
MOV P1, #00H
SETB P1.0
ACALL DELAY
INC R1
CJNE R1, #08H, LEFT_LOOP
ACALL DELAY
CASE2_RIGHT:
MOV R1, #07H
RIGHT_LOOP:
MOV P1, #00H
SETB P1.7
ACALL DELAY
DJNZ R1, RIGHT_LOOP
ACALL DELAY
CASE3_BLINK_SETS:

BLINK_LOOP:
MOV P1, #F0H
ACALL DELAY
```

```
MOV P1, #0FH
ACALL DELAY
DJNZ R0, BLINK_LOOP
CASE4_BLINK_ALL:
BLINK_ALL_LOOP:
MOV P1, #00H
ACALL DELAY
MOV P1, #FFH
ACALL DELAY
DJNZ R0, BLINK_ALL_LOOP
JMP START
DELAY:
MOV R2, #255
DELAY_OUTER:
MOV R3, #255
DELAY_INNER:
DJNZ R3, DELAY_INNER
DJNZ R2, DELAY_OUTER
RET
END
```

OUTPUT:





Experiment #7

Demonstration and practice of Dot Matrix LED Control and display different patterns.

TASK#1:

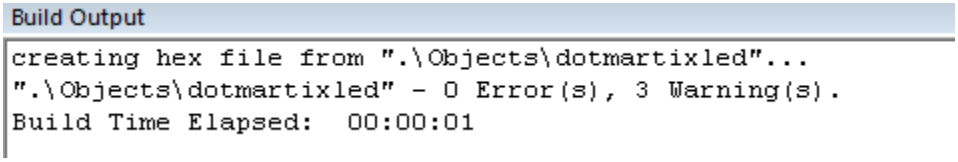
Display the example in the manual and analyze the output.

CODE:

```
ORG 0000H
MOV P1, #0FFH
MOV P2, #00H
A_PATTERN: DB 0x18, 0x3C, 0x66, 0xC3, 0xFF, 0xC3, 0xC3, 0xC3
MAIN:          ; Main loop to continuously display the letter
   MOV R0, #A_PATTERN
DISPLAY:
   MOV R1, #08H
DISPLAY_ROW:
   MOV A, @R0
   MOV P1, A
   MOV A, R1
   CPL A
   MOV P2, A
   ACALL DELAY
   INC R0
   MOV A, R1
   RL A
   MOV R1, A
   DJNZ R1, DISPLAY_ROW
   SJMP MAIN
```

```
DELAY:
   MOV R2, #255
DELAY_LOOP1:
   DJNZ R2, DELAY_LOOP1
   MOV R2, #255
DELAY_LOOP2:
   DJNZ R2, DELAY_LOOP2
   RET
END
```

OUTPUT:



TASK#2:

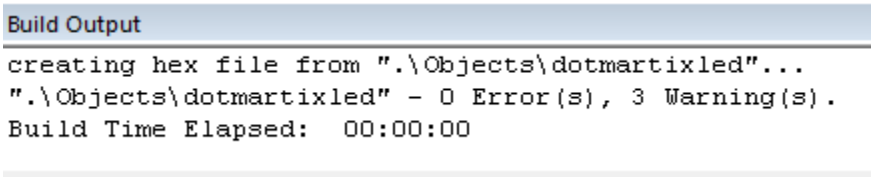
Write a program which display the alphabet A on Dot Matrix LED.

CODE:

```
ORG 0000H
PATTERN: DB 3CH, 66H, 66H, 7EH, 66H, 66H, 66H, 00H
ROW_SELECT: DB 0FEH, 0FDH, 0FBH, 0F7H, 0EFH, 0DFH, 0BFH, 0FFH
MOV DPTR, #PATTERN
MOV P0, #0FFH
MOV P2, #0FFH        ; Clear Port 2 (connected to rows)
MAIN_LOOP:
```

```
    MOV R0, #08      ; Loop counter for 8 rows
DISPLAY_LOOP:
    MOVX A, @DPTR    ; Load row data from pattern
    MOV P2, A        ; Output row data to Port 2 (74LS245 for rows)
    MOV A, R0
    DEC A            ; Convert R0 to match row index (0-7)
    MOVC A, @A+ROW_SELECT ; Get row selection value from ROW_SELECT
    MOV P0, A        ; Output to Port 0 (ULN2803 for columns)
    ACALL DELAY      ; Call delay for stable display
    INC DPTR         ; Move to next row in the pattern
    DJNZ R0, DISPLAY_LOOP ; Repeat for all rows
    MOV DPTR, #PATTERN ; Reset DPTR to start of pattern
    SJMP MAIN_LOOP     ; Repeat the main loop indefinitely
; Simple delay subroutine
DELAY:
    MOV R1, #0FFH
    MOV R2, #0FFH
DELAY_LOOP:
    DJNZ R2, DELAY_LOOP
    DJNZ R1, DELAY_LOOP
    RET
END
```

OUTPUT:



TASK#3:

```
ORG 0000H
MOV DPTR, #L_A
CALL DISPLAY
MOV DPTR, #L_B
CALL DISPLAY
MOV DPTR, #L_C
CALL DISPLAY
SJMP $
DISPLAY:
MOV R0, #8
DISPLAY_LOOP:
MOVX A, @DPTR
MOV P1, A
ACALL DELAY
INC DPTR
DJNZ R0, DISPLAY_LOOP
CALL DELAY_500MS
RET
L_A: DB 0x3C, 0x66, 0xC3, 0xFF, 0xC3, 0xC3, 0xC3, 0x00
L_B: DB 0xFC, 0x66, 0x66, 0xFC, 0x66, 0x66, 0xFC, 0x00
L_C: DB 0x3C, 0x66, 0xC0, 0xC0, 0xC0, 0x66, 0x3C, 0x00
DELAY_500MS:
MOV R1, #50 D1: MOV R2, #255
D2: MOV R3, #255
DJNZ R3, D2
DJNZ R2, D1

DJNZ R1, DELAY_500MS
```

```
RET
DELAY:
MOV R4, #100
D3: DJNZ R4, D3
RET
END
```

OUTPUT:

```
Build Output
Program Size: data=9.0 xdata=0 code=72
".\Objects\dotmartixledaplahabet" - 0 Error(s), 3 Warning(s).
Build Time Elapsed:  00:00:00
```

```
Build Output
creating hex file from ".\Objects\dotmartixledaplahabet"...
".\Objects\dotmartixledaplahabet" - 0 Error(s), 3 Warning(s).
Build Time Elapsed:  00:00:01
```